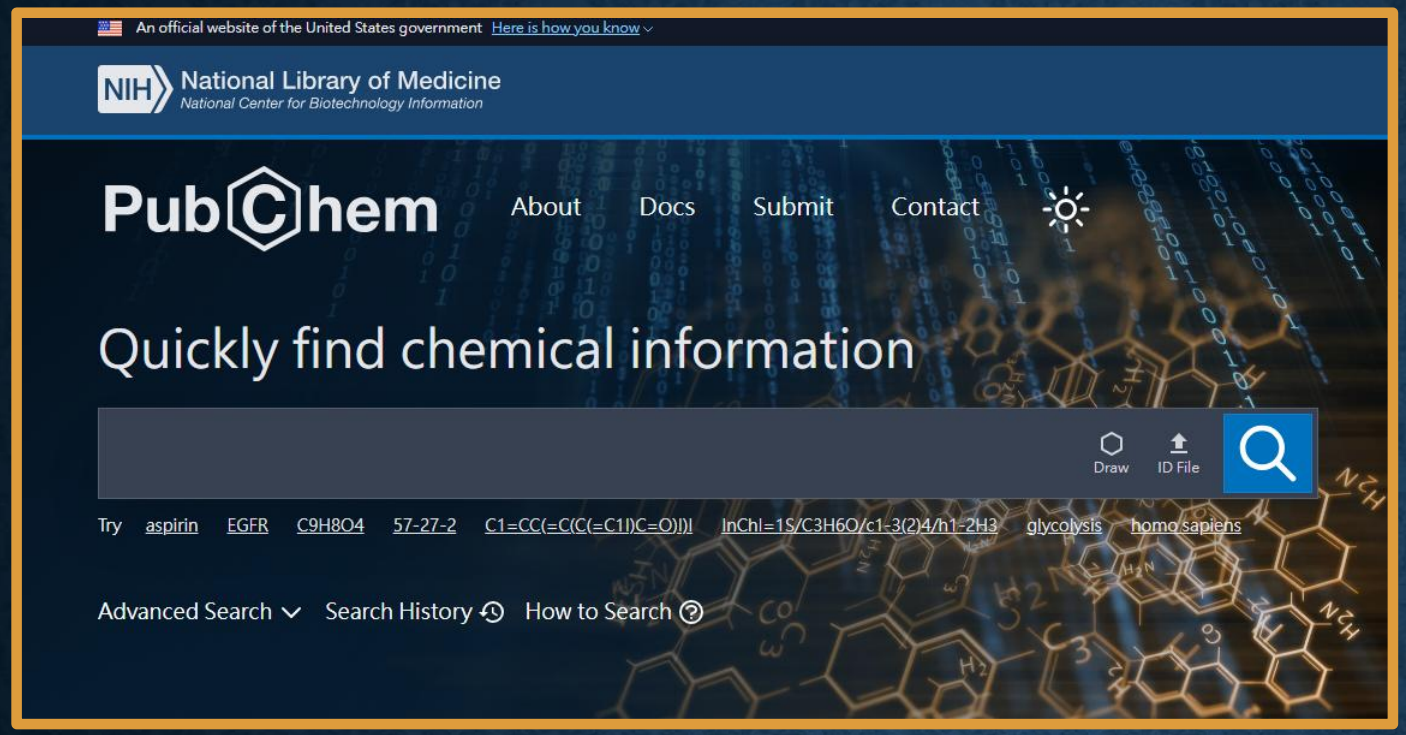
MD & CADD DEMONSTRATIONS

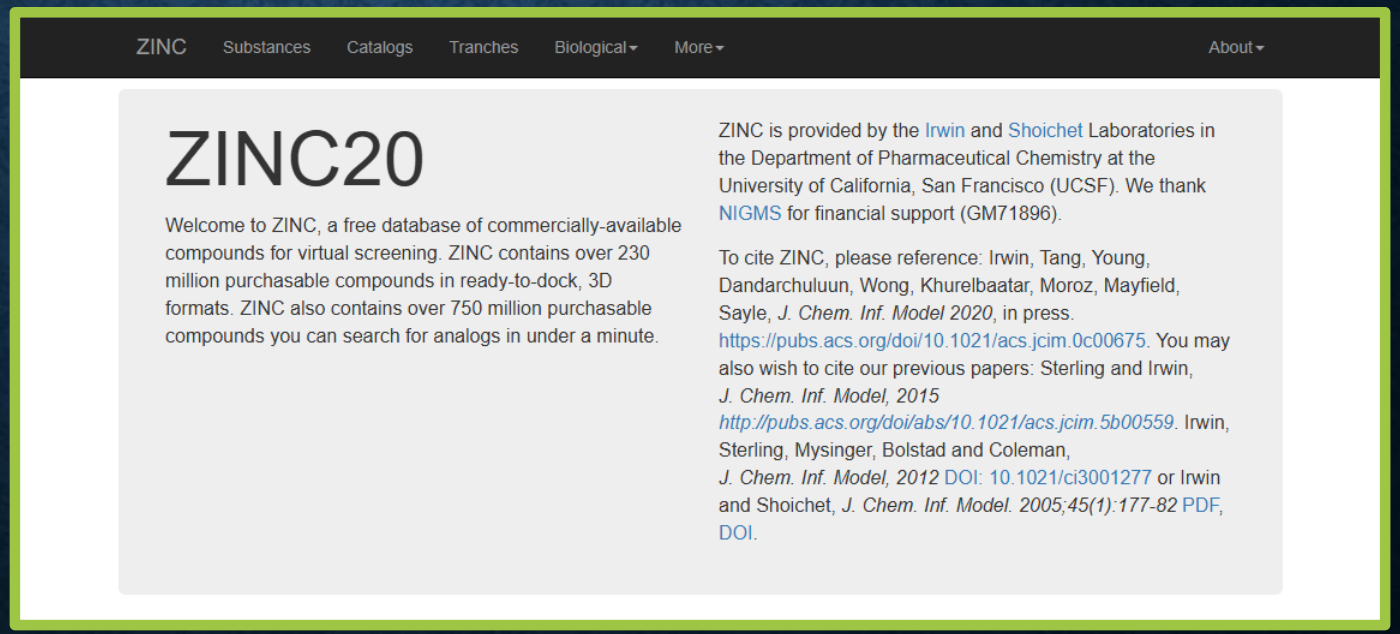
OBTAINING SMALL MOLECULE STRUCTURES

ACQUISITION STRATEGIES: DATABASE VS. MODELING

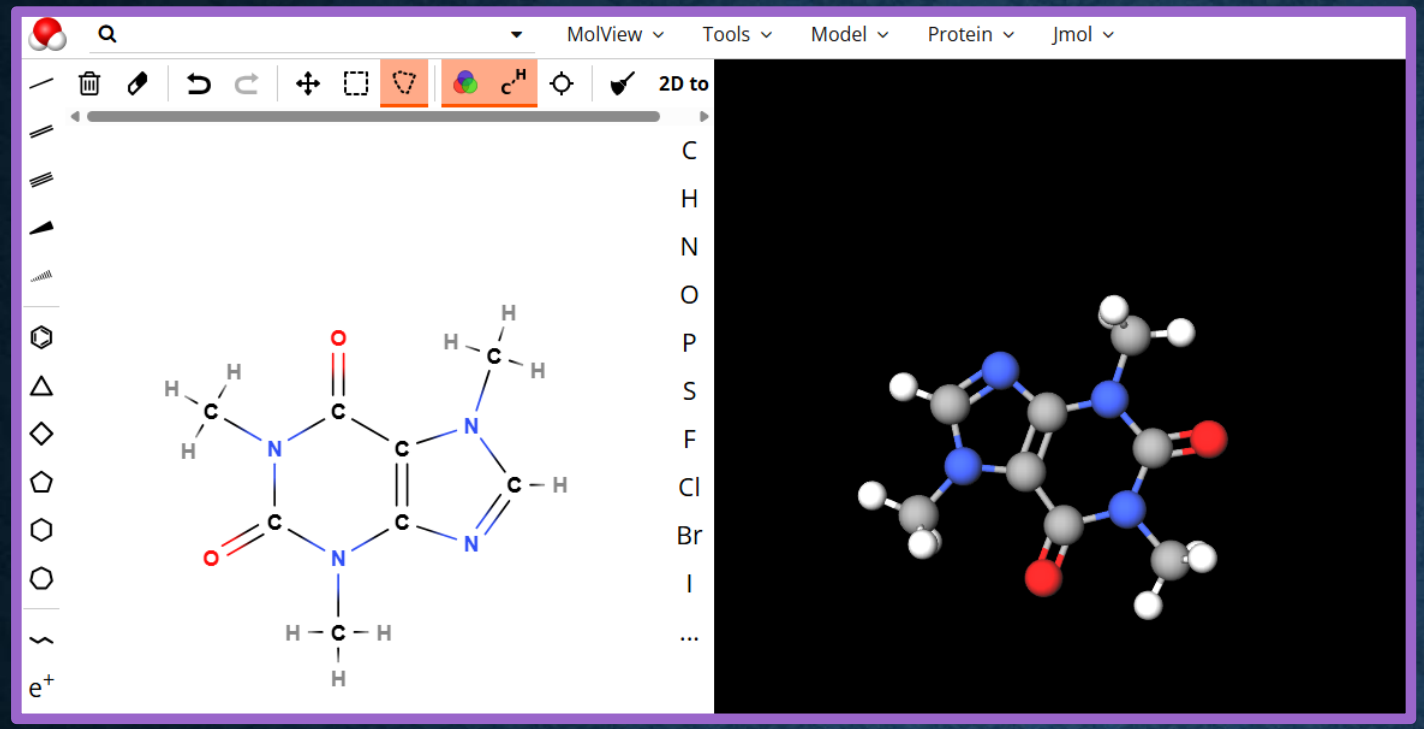
- Database Retrieval
  - PubChem: The world's largest free chemical database (NIH). (https://pubchem.ncbi.nlm.nih.gov/)
  - ZINC20: Specialized in commercially available compounds for docking. (https://zinc.docking.org/)
- Structural Drawing & Modeling
  - MolView: An intuitive Open-Source web tool for 3D visualization (online tool). (https://molview.org/)
  - Avogadro & Avogadro 2: Open-source advanced molecule editor and visualizer (local software). (https://avogadro.cc/ & https://www.openchemistry.org/projects/avogadro2/)
  - ChemDraw: The industry standard for professional 2D/3D rendering (local software). (https://revvitysignals.com/products/research/chemdraw)



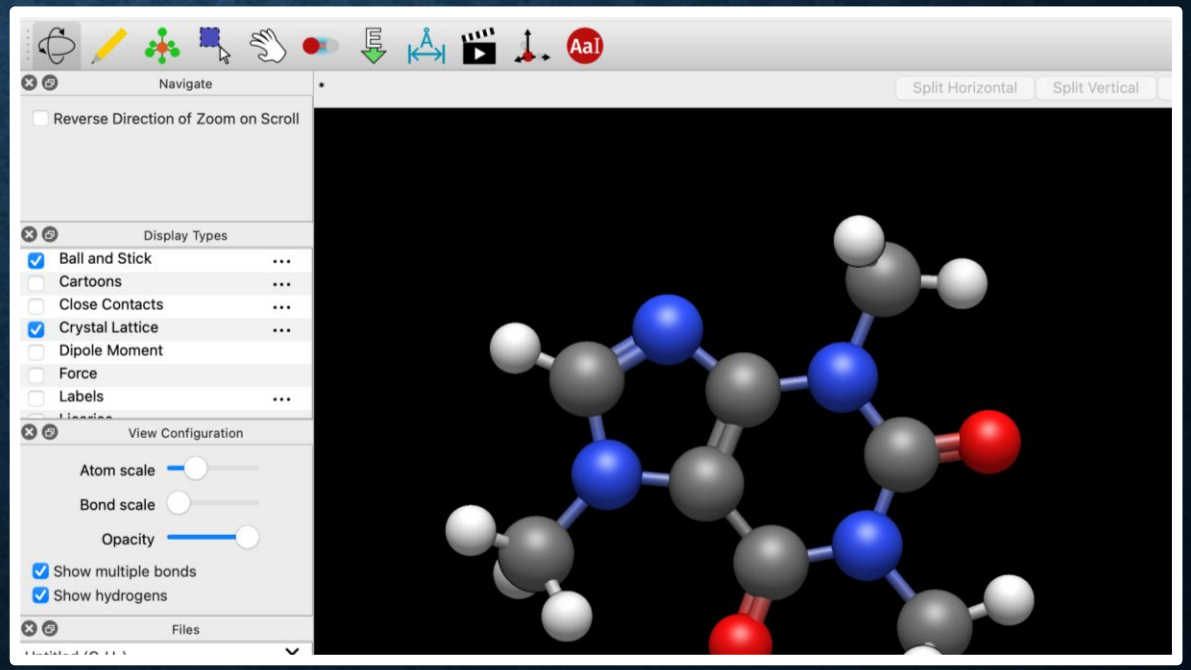
PubChem



ZINC20



MolView



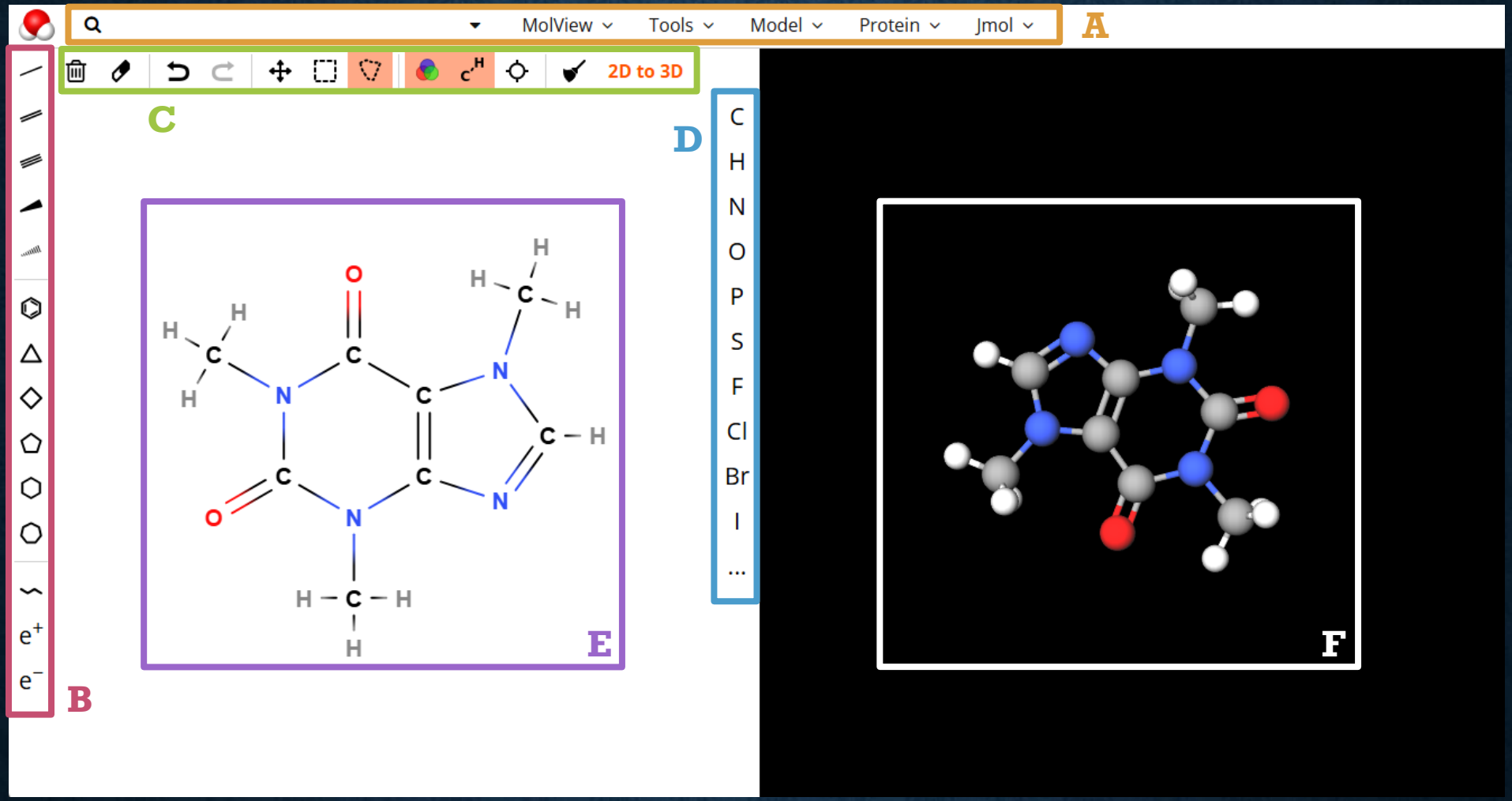
Avogadro

SEARCHING PUBCHEM

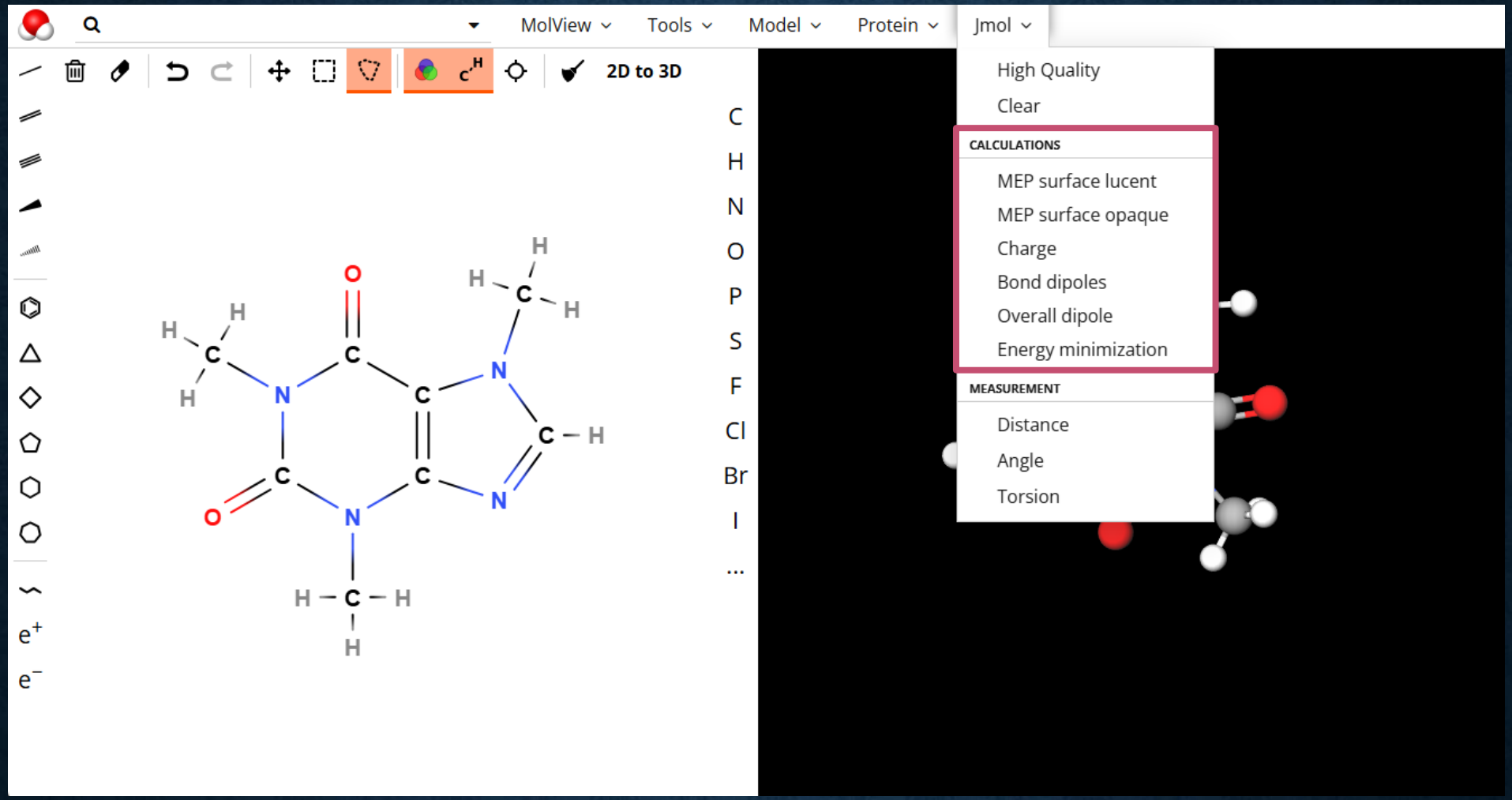
- Basic Text Search
  - Common Names: Just type the name (e.g., "Caffeine" or "Aspirin").
  - IUPAC Name: Using the formal systematic chemical name.
  - CAS Number: The unique "ID card" number for chemicals (e.g., 50-78-2 for Aspirin).
- Line Notations
  - SMILES String: A short text string that represents a chemical structure. (e.g., CC(=O)OC1=CC=CC=C1C(=O)O for Aspirin)
  - InChI / InChIKey: A unique, non-proprietary digital signature for any molecule. (e.g., InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12) for Aspirin)

MOLVIEW
ALL-IN-ONE ONLINE 3D MODELER

1. High Accessibility
   - No Installation Required – Fully functional on any web browser and device.
   - Database Integration – Instant search and import from PubChem and other databases.
2. Smart 2D-to-3D Workflow
   - One-Click Conversion – Seamlessly transform flat sketches into spatial 3D models.
   - Real-Time Geometry – Interactive rotation, zooming, and structural measurements.
3. Advanced Research Support
   - Energy Minimization – Built-in Jmol engine for realistic molecular geometry.
   - Versatile Export – Save structure in .mol format for further analysis.
   - Spectral Data – Access IR, Mass, and NMR spectra where available via NIST/PubChem integration.

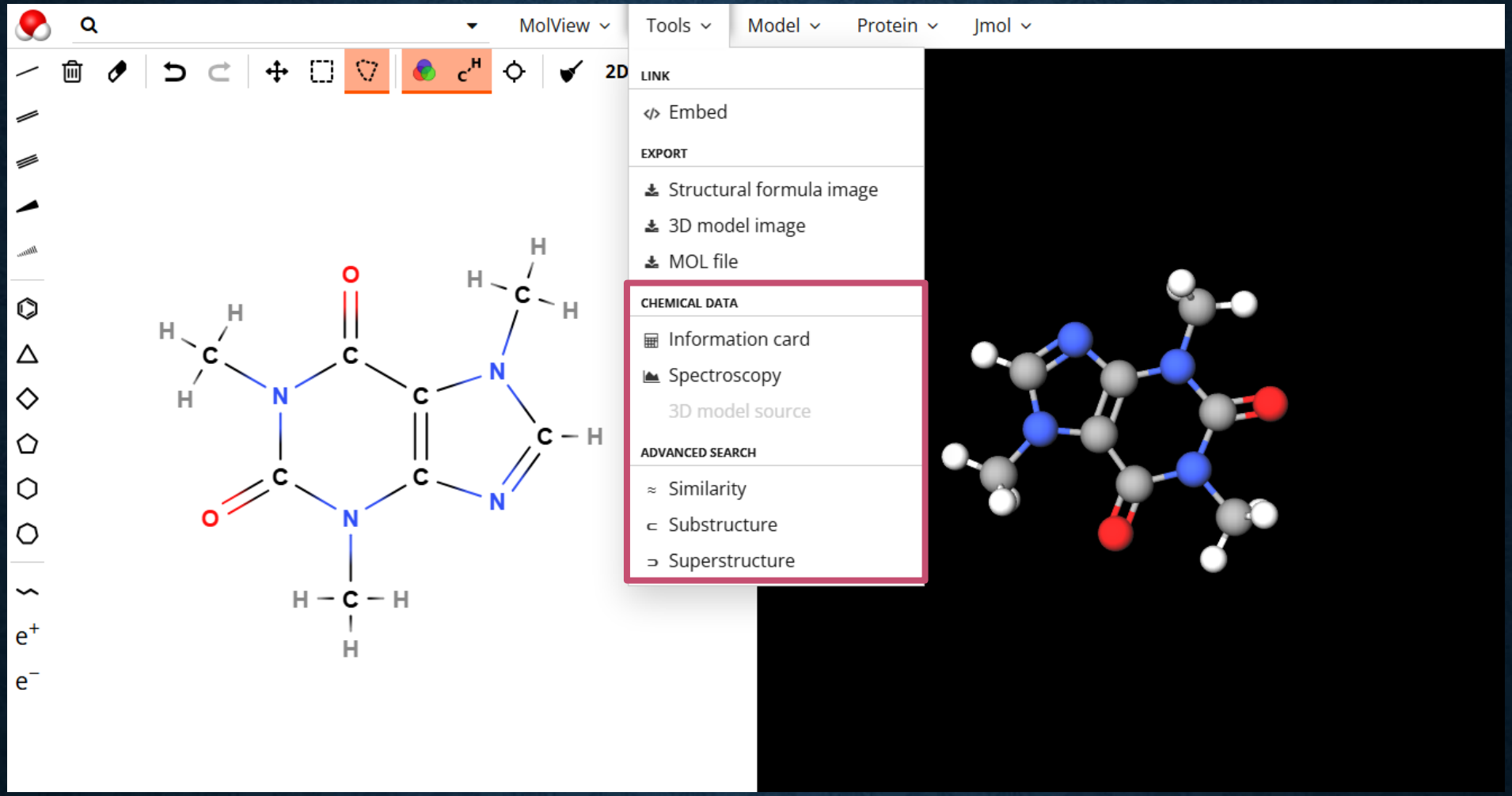


A. Main Menu – Advanced tools, modeling modes, and file export.
B. Drawing Tools – Tools for bonds, rings, and charging.
C. Action Toolbar – Undo/redo, clear canvas, and 2D to 3D button.
D. Element List – Quick access to select different atoms.
E. 2D Canvas – Workspace for sketching 2D molecular structures.
F. 3D Viewer – Rotate, zoom, and visualize the 3D model.



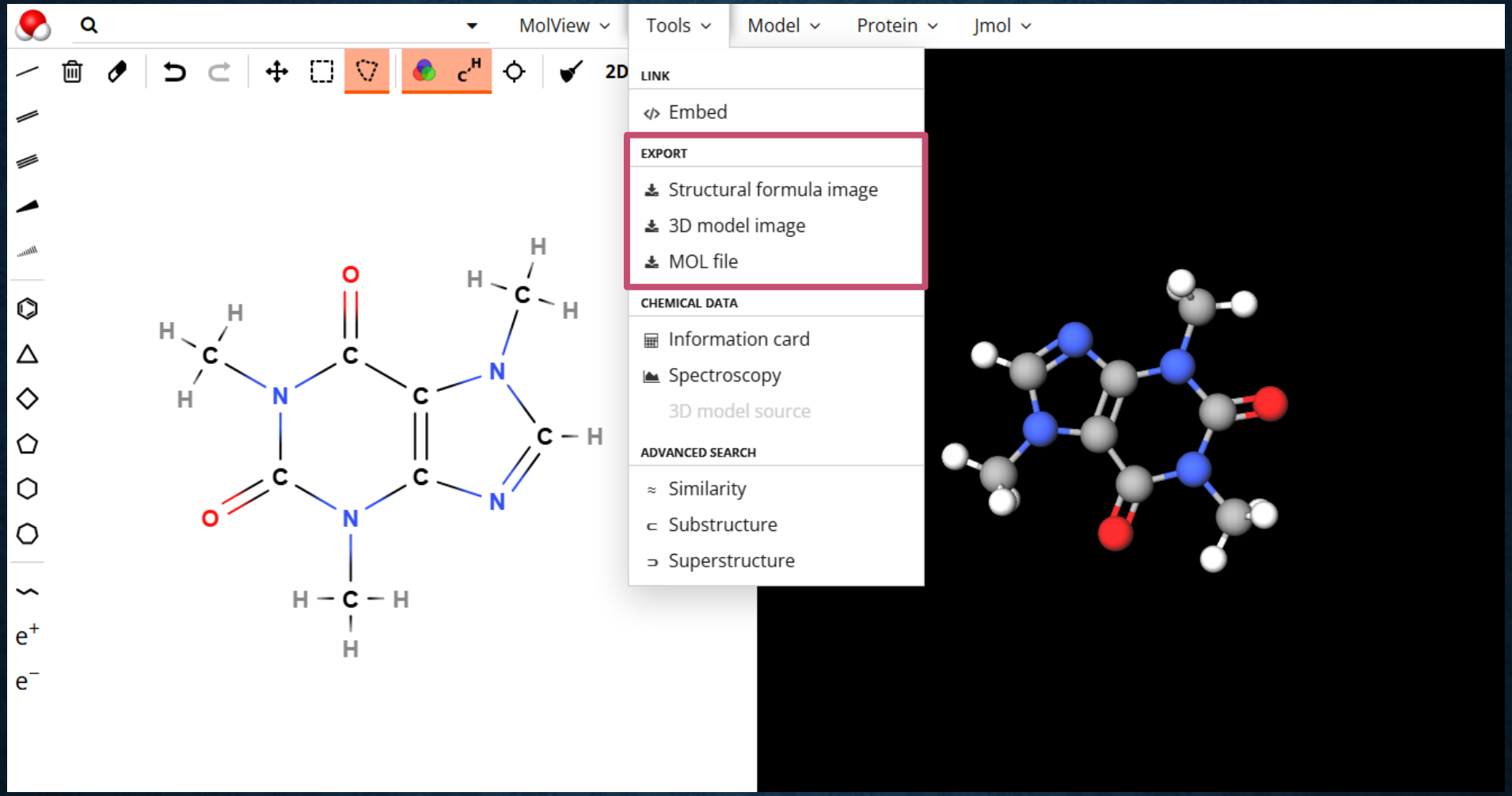
Computational Features in MolView (Jmol)

- Energy Minimization – Optimizes 3D structure using **MMFF94** to find most stable conformation.
- Charge – Calculates atomic partial charges to show election distribution.
- MEP Surface – Maps electrostatic potential onto the surface to identify polarity.
- Dipole Moments – Visualizes the molecular polarity and its direction.



Chemical Data & Advanced Search (Data Retrieval via PubChem)

- Information card – Displays the molecules identity (Name, Weight, CAS).
- Spectroscopy - Accesses experimental or predicted spectra (IR, MS, NMR).
- Similarity – Finds molecules with a similar chemical fingerprint.
- Substructure – Searches for larger molecules containing this fragment.
- Superstructure – Identifies simpler parent structures or frameworks.



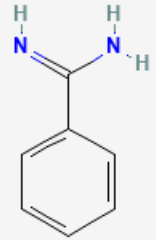
Export (Data Output)

- Structural formula images – Saves the 2D sketch as a high-quality image.
- 3D model image – Captures the 3D visualization as a static picture.
- MOL file – Downloads the .mol file containing 3D XYZ coordinates.

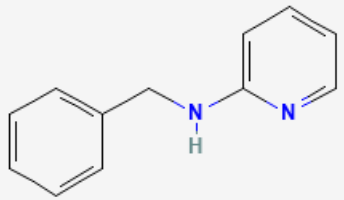
FROM 2D SKETCH TO 3D MODEL

1. Sketching
   - Select Atoms – Choose elements (C, O, N etc.) from the Element List.
   - Draw Bonds – Use Drawing Tools to click and drag on the 2D Canvas to create single, double, or triple bonds.
   - Add Rings – Quickly insert benzene or cyclohexane rings from the shortcut icons in Drawing Tools.
2. Conversion
   - 2D to 3D – Click the "2D to 3D" button in the Action Toolbar.
   - Visualize – The 3D model will instantly appear in the 3D Viewer. You can rotate and zoom to inspect the geometry.
3. Optimization (Energy Minimization)
   - Access Jmol – Go to Jmol in the Main Menu.
   - Refine Structure: – Select "Energy Minimization".
4. Exporting (Saving the Result)
   - Access Tools – Go to Tools in the Main Menu.
   - Download File – Select "Export" → "MOL File (.mol)".

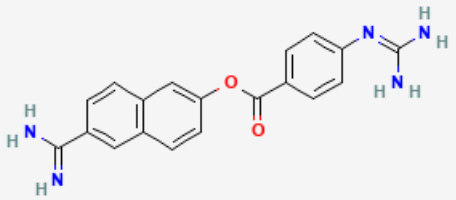
MOLECULAR SKETCHING



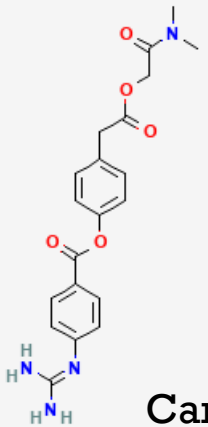
Benzamidine



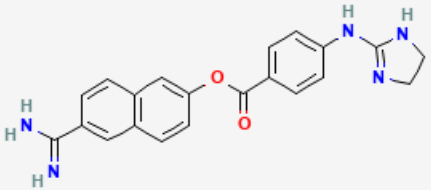
BMAP



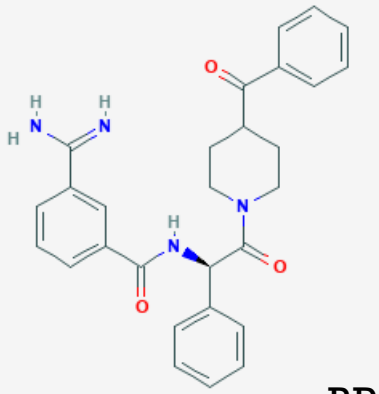
Nafamostat



Camostat



Sepimostat



BPO

SUPPLEMENTARY INFORMATION

SMILES: THE DIGITAL LANGUAGE FOR CHEMICAL AI

1. Why It Matters: The Bridge to Chemical Structure & AI
   - Sequential Representation – Converts complex 3D molecular topologies into simple 1D text strings.
   - LLM Compatible – Enables Large Language Models (LLMs) like GPT or Transformers to "read" and "generate" molecules just like human language.
   - High Efficiency – The fastest and most compact format for processing millions of compounds in drug discovery pipelines.
2. Core Concept: Compact & Highly Informative
   - Atoms – Represented by element symbols (e.g., C, N, O).
   - Bonding Rules – Single bonds are implied; use = or # for multiple bonds.
   - Molecular Topology – Parentheses "( )" for branches and numbers for ring closures.

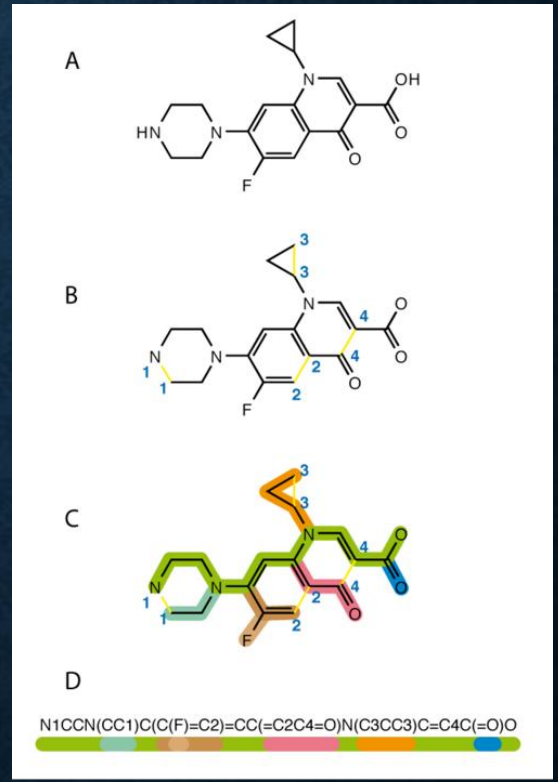


Figure Source: https://commons.wikimedia.org/wiki/File:SMILES.png

A GUIDE TO PROTEIN STRUCTURE AND MODELING

THE CORE OF STRUCTURAL BIOLOGY
3D COORDINATES OF BIOMACROMOLECULES

- Experimental Method to Resolve 3D Structures
  - X-ray Crystallography
    Uses high-energy X-ray diffraction to interact with electron clouds, requiring proteins to be highly purified and crystallized into an ordered 3D lattice to amplify signals.
  - NMR Spectroscopy
    Detects magnetic properties (spins) of specific isotopes ^{1}H, ^{13}C, ^{15}N) in a strong magnetic field, allowing structure determination in aqueous solution to capture protein dynamics.
  - Cryo-Electron Microscopy (Cryo-EM)
    Uses an **accelerated electron beam** to directly image molecules flash-frozen in **vitreous ice**, preserving their native state without the need for crystallization.

STRUCTURAL BIOLOGY REPOSITORIES: ORGANIZATIONS AND PORTALS

- wwPDB (Worldwide Protein Data Bank)
  - The international organization that maintains the unified global archive and sets data standards.
  - It dictates the file formats (like PDB or PDBx/mmCIF) to ensure data can be read by all software (VMD, PyMOL, ChimeraX).
- RCSB PDB (rcsb.org)
  - The primary web portal (based in the USA) used by researchers to search, visualize, and download data.
  - Where scientists upload their 3D (x, y, z) coordinates after an experiment.
  - Each structure is assigned a PDB ID and undergoes a rigorous Validation Report process.
  - Regional Mirrors & Partners: PDBe (European) and PDBj (Japan).

RCSB Protein Data Bank (https://www.rcsb.org/)



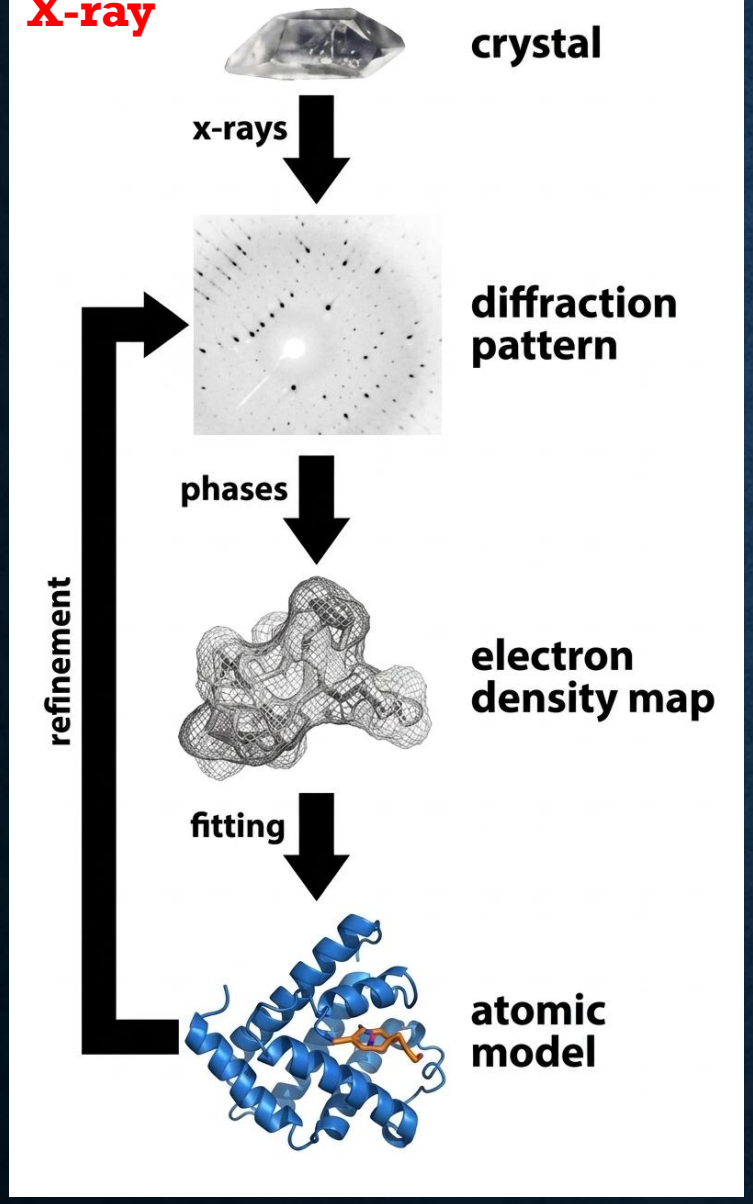
PDBe (https://www.ebi.ac.uk/pdbe/)



PDBj (https://pdbj.org/)

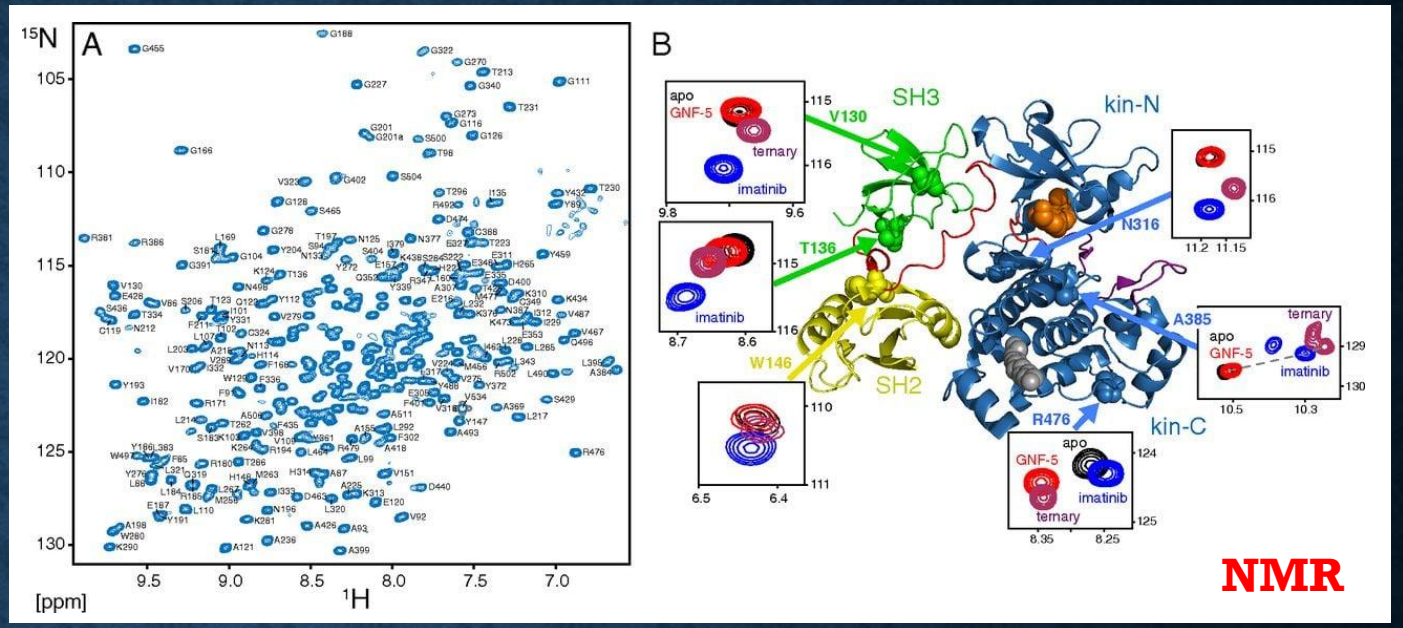


X-ray



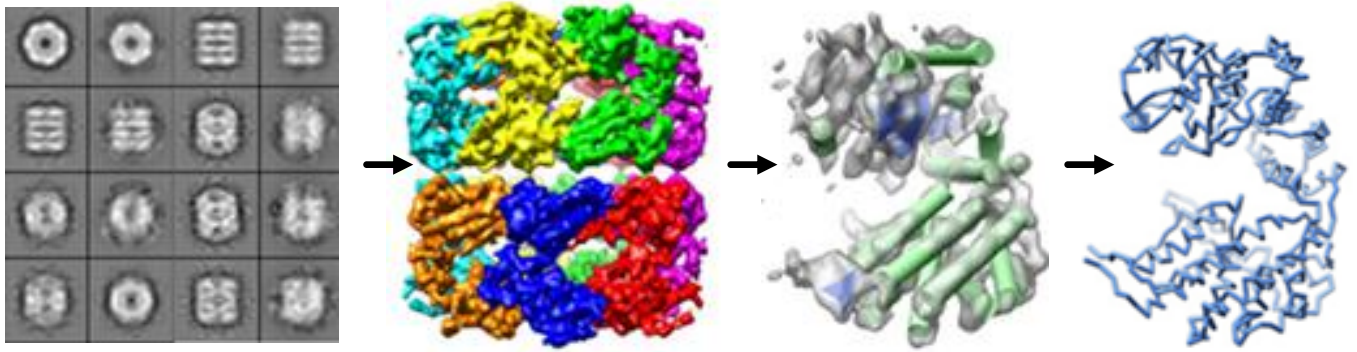
Source: https://www.nsrrc.org.tw/chinese/Industrial.aspx

NMR

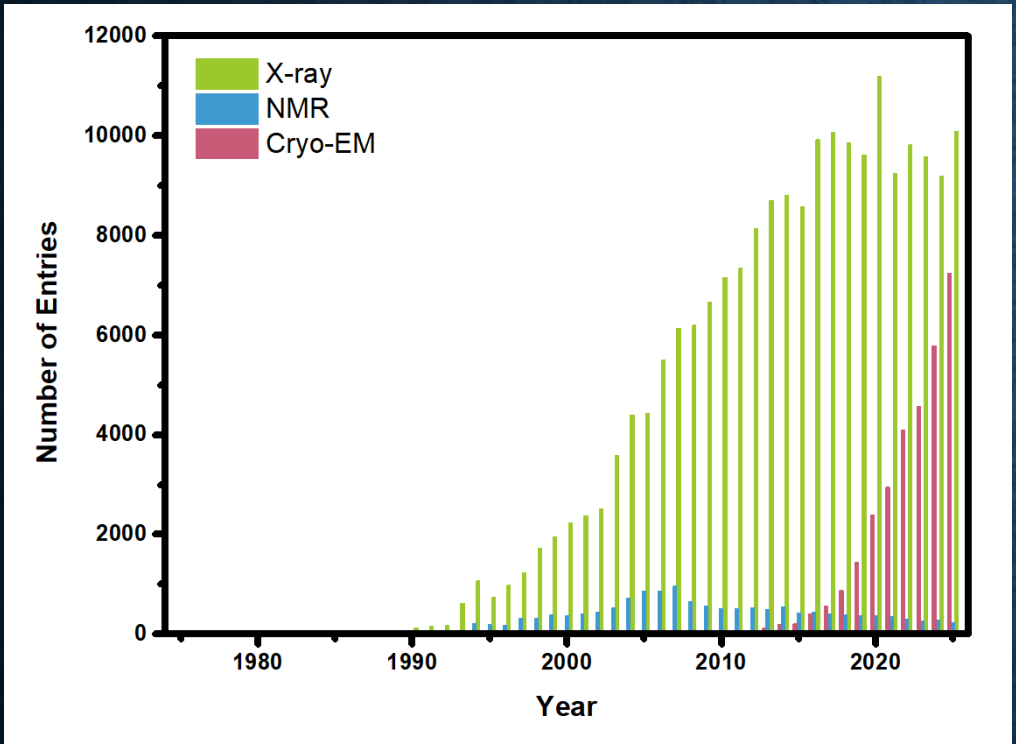


Source: https://www.pnas.org/doi/10.1073/pnas.1314712110

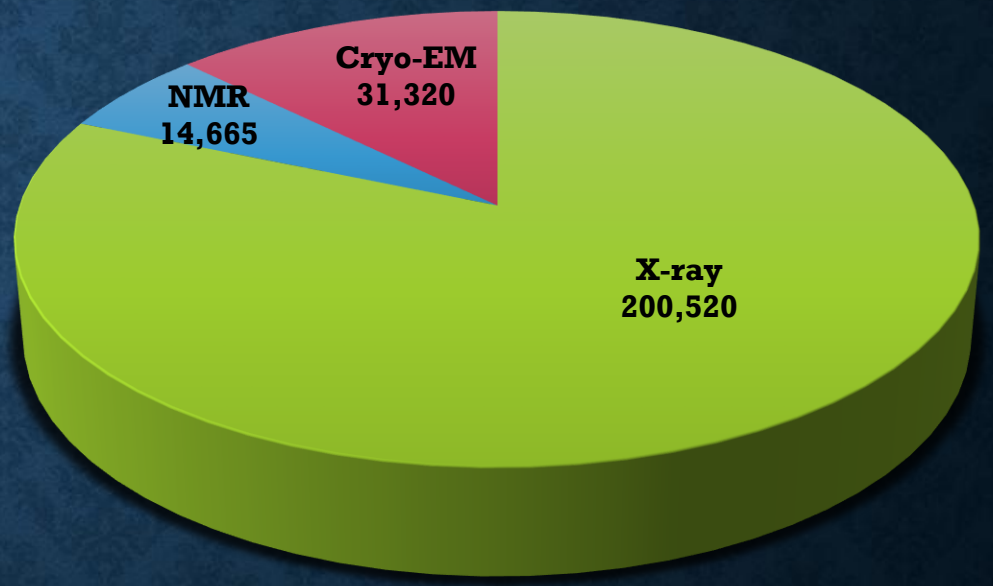
Cryo-EM



Source: https://investigator.tw/



Evolution of Structural Determination Techniques



Distribution of Structures by Experimental Method

HANDS-ON: RETRIEVING PROTEIN STRUCTURES

1. Search the Protein Structures
   - Go to the Protein Data Bank (https://www.rcsb.org/).
   - Search for "**Trypsin inhibitor complex (BPO)**" and navigate to the page for PDB ID 1EB2.
2. Download the Structures
   - Click the dropdown menu on the "Download Files" buttons.
   - Download both of the following formats: PDBx/mmCIF and Legacy PDB.
3. Inspect the Content
   - Open both files using a Text Editor (e.g., Notepad, TextEdit, or VS Code).
   - Observe: Browse the text to find the coordinate data and see how the two formats differ in structure.

RCSB PDB | Deposit | Search | Visualize | Analyze | Download | Learn | About | Careers | COVID-19 | Help | Contact us | MyPDB

RCSB PDB PROTEIN DATA BANK

252,476 Structures from the PDB archive

1,068,577 Computed Structure Models (CSM)

Enter search term(s), Ligand ID or sequence | Include CSM

Advanced Search | Chemical Search | Browse Annotations | Help

PDB-101 | wwPDB | EMDataResource | NAKB | wwPDB Foundation | PDB-IHM

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Versions

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Validation Report | Ligand Interaction (BPO)

Global Symmetry: Asymmetric - C1

Global Stoichiometry: Monomer - A1

Display Files | Download Files | Data API

1EB2 | pdb_00001eb2

Trypsin inhibitor complex (BPO)

PDB DOI: https://doi.org/10.2210/pdb1EB2/pdb

Classification: HYDROLASE
Organism(s): Bos taurus
Mutation(s): No

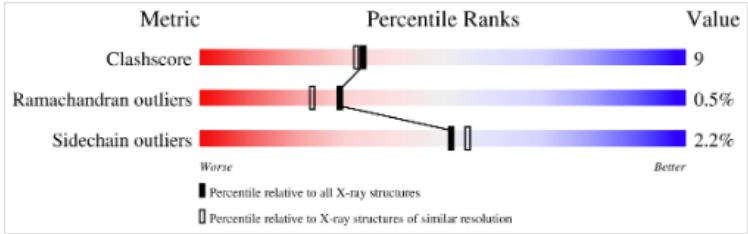
Deposited: 2001-07-18 **Released:** 2002-02-11
Deposition Author(s): Wilkinson, K.W., Young, S.C., Liebeschuetz, J.W., Brady, R.L.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.00 Å
R-Value Free:
0.244 (Depositor)
R-Value Work:
0.178 (Depositor)
R-Value Observed:
0.178 (Depositor)

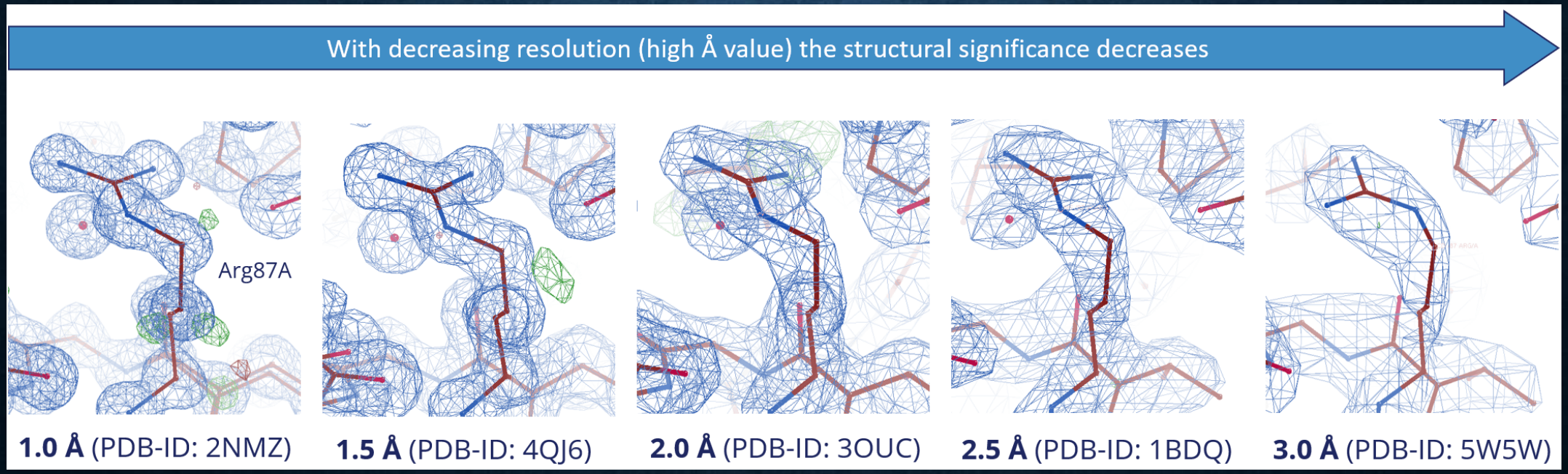
Starting Model: experimental
View more details

wwPDB Validation | 3D Report | Full Report



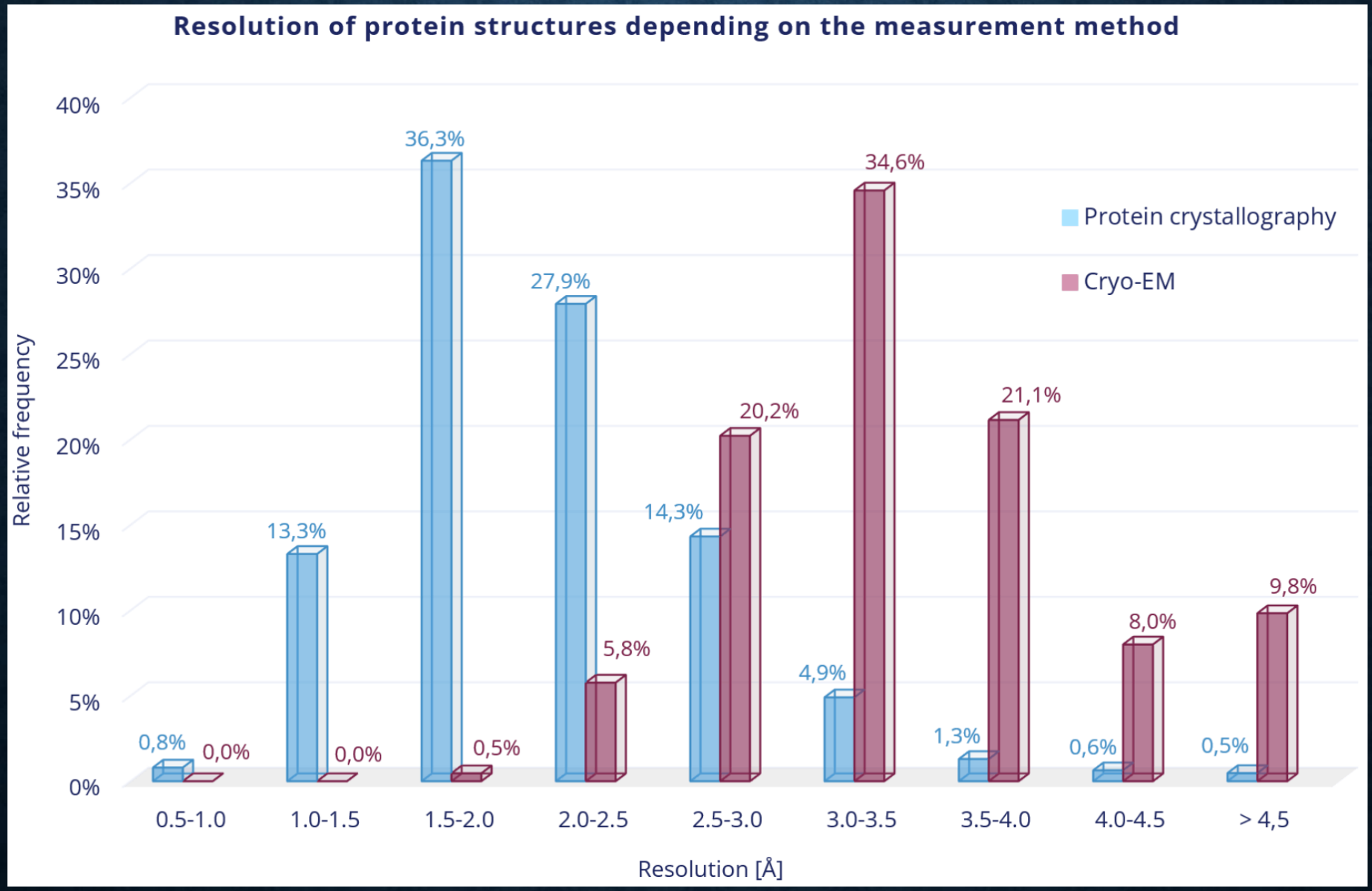
Understanding Resolution in Structural Biology (Only for X-ray and Cryo-EM)

Resolution (Å)	What we can actually see	Quality Level
< 1.2 Å	Individual atoms, Hydrogen atoms.	Atomic
1.5 - 2.5 Å	Side chains, water molecules, drug ligands	Excellent
3.0 - 4.0 Å	Main backbone fold, but side chains are shaky.	Fair
> 5.0 Å	Global shape, alpha-helices as tubes.	Low



Source: https://stbio.de/aufloesung_en.html

Understanding Resolution in Structural Biology (Only for X-ray and Cryo-EM)



Source: https://stbio.de/aufloesung_en.html

THE POWER OF MOLECULAR VISUALIZATION

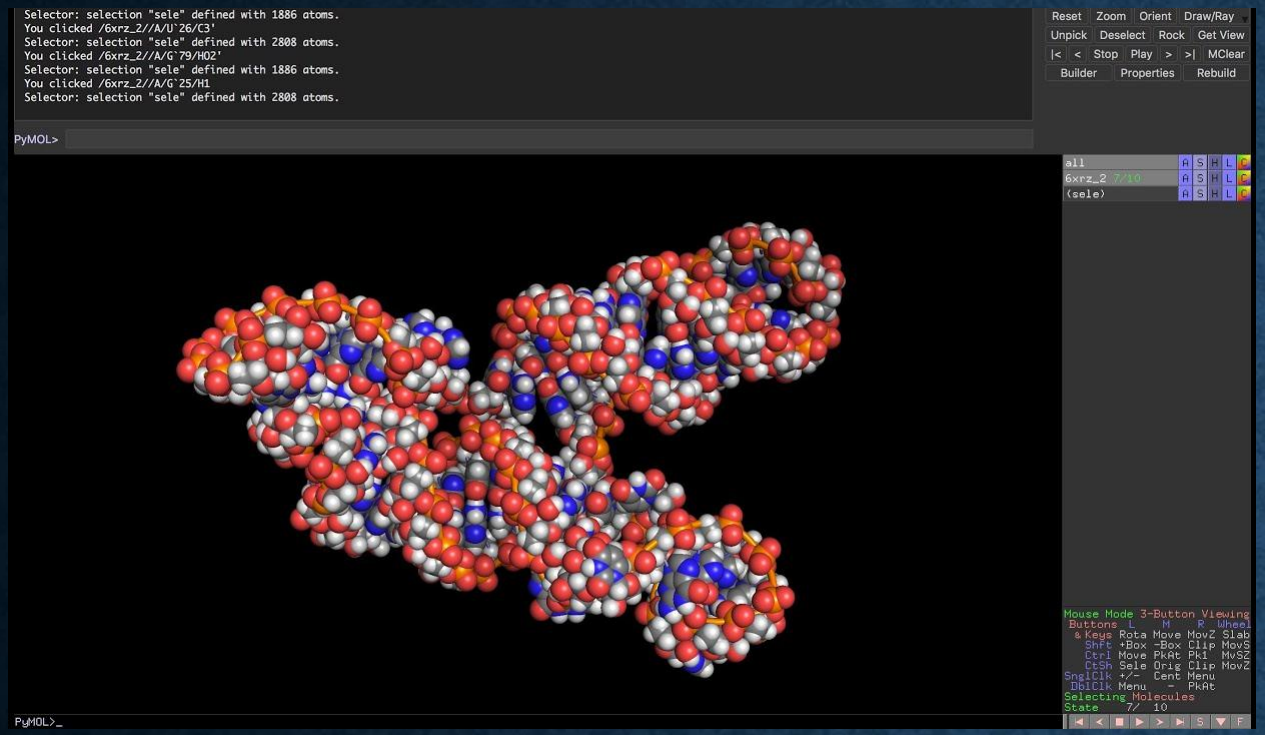
- Bridge the Gap: Text to Vision
  - The Reality: Structural files are just lists of x, y, z coordinates.
  - The Solution: Visualization turns "data" into a "3D Molecular Microscope.“
- Why Visualization Matters?
  - **Insight: See drug-binding pockets and active sites directly.**
  - **Dynamics:** Watch proteins move and function in real-time.

VMD VS. PYMOL

- VMD (Visual Molecular Dynamics)
  - Focus: Large systems & Dynamics (Movies).
  - Pro: Free, open-source, and best for NAMD simulation.
- PyMOL
  - **Focus: Publication-quality** static images.
  - Pro: Superior default aesthetics; easy for "pretty" pictures.

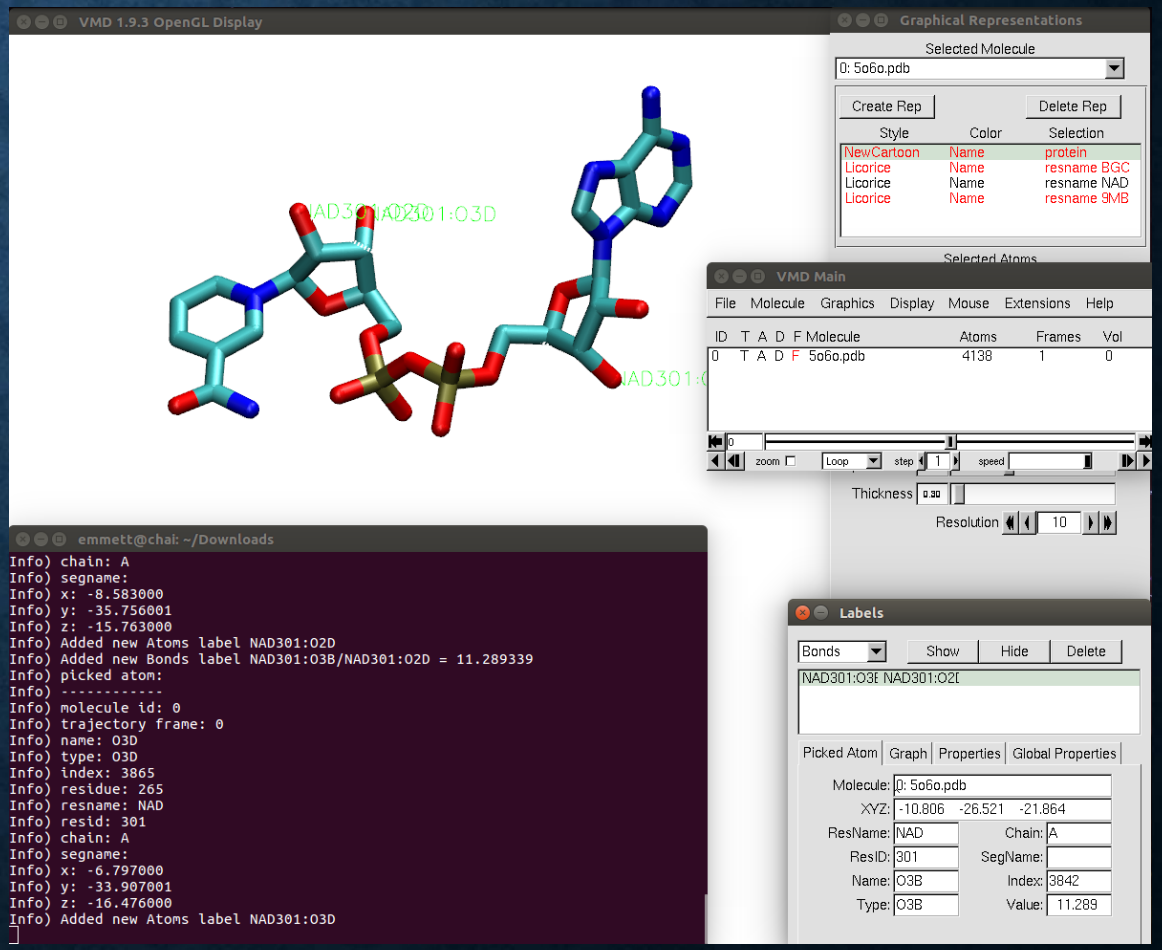
Our Choice for Today: VMD

1. Zero Cost: No licensing hurdles for students.
2. Simulation Ready: If you want to run simulations (NAMD), you must know VMD.


Selector: selection "sele" defined with 1886 atoms.
You clicked /6xrz_2//A/U`26/C3'
Selector: selection "sele" defined with 2808 atoms.
You clicked /6xrz_2//A/G`79/HO2'
Selector: selection "sele" defined with 1886 atoms.
You clicked /6xrz_2//A/G`25/H1
Selector: selection "sele" defined with 2808 atoms.
Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
Stop Play MClear
Builder Properties Rebuild
PyMOL>
all
6xrz_2 7/10
(sele)
Mouse Mode 3-Button Viewing
Selecting Molecules
State 7/ 10
PyMOL>_

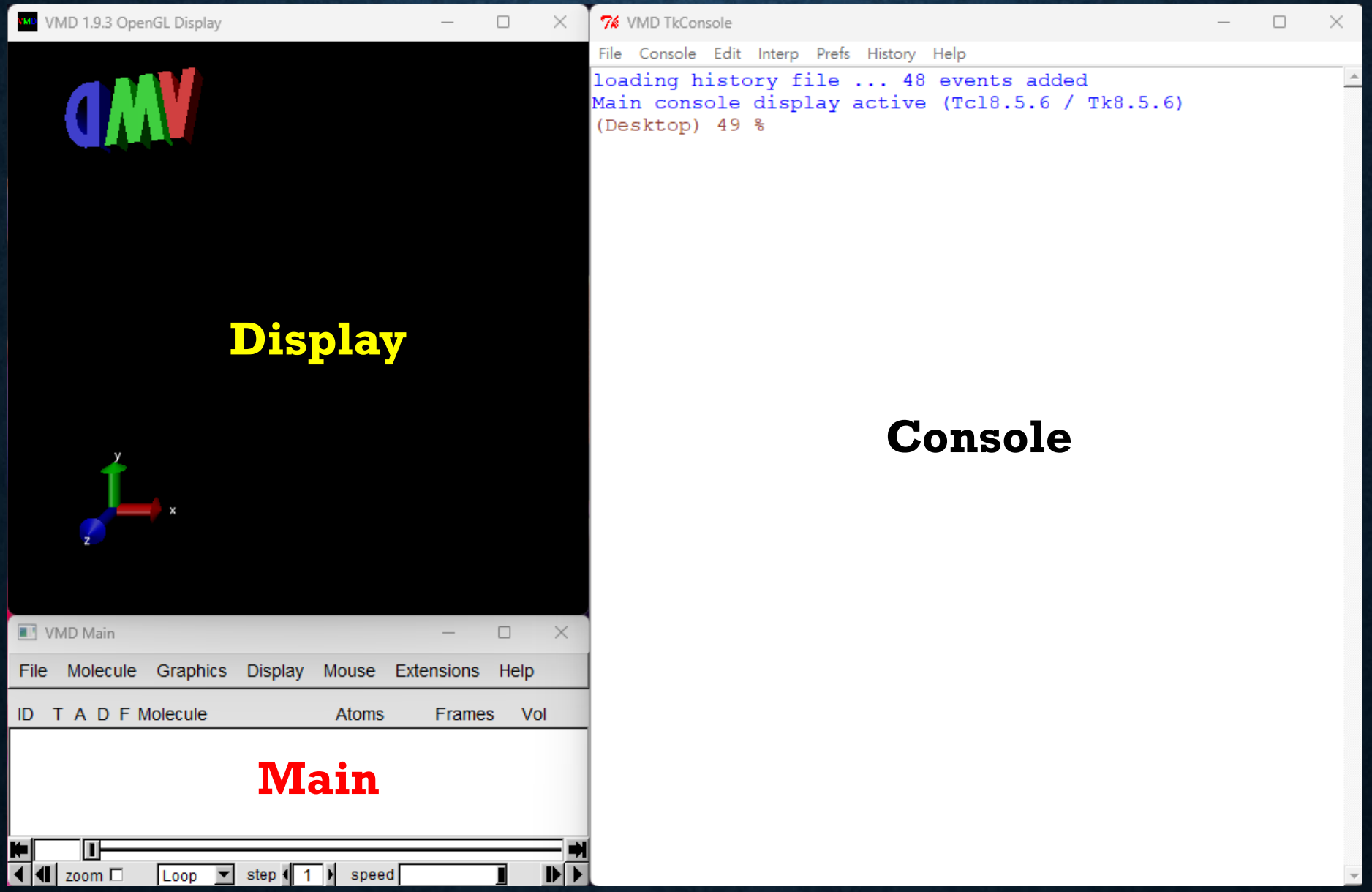
PyMOL

VMD


VMD 1.9.3 OpenGL Display
Graphical Representations
Selected Molecule
0: 5o6o.pdb
Create Rep
Delete Rep
Style Color Selection
NewCartoon Name protein
Licorice Name resname BGC
Licorice Name resname NAD
Licorice Name resname 9MB
Selected Atoms
VMD Main
File Molecule Graphics Display Mouse Extensions Help
ID T A D F Molecule Atoms Frames Vol
0 T A D F 5o6o.pdb 4138 1 0
Thickness
Resolution
10
emmett@chai: ~/Downloads
Info) chain: A
Info) segname:
Info) x: -8.583000
Info) y: -35.756001
Info) z: -15.763000
Info) Added new Atoms label NAD301:O2D
Info) Added new Bonds label NAD301:O3B/NAD301:O2D = 11.289339
Info) picked atom:
Info) ------------
Info) molecule id: 0
Info) trajectory frame: 0
Info) name: O3D
Info) type: O3D
Info) index: 3865
Info) residue: 265
Info) resname: NAD
Info) resid: 301
Info) chain: A
Info) segname:
Info) x: -6.797000
Info) y: -33.907001
Info) z: -16.476000
Info) Added new Atoms label NAD301:O3D
Labels
Bonds
Show
Hide
Delete
Picked Atom
Graph
Properties
Global Properties
Molecule: 0: 5o6o.pdb
XYZ: -10.806 -26.521 -21.864
ResName: NAD
Chain: A
ResID: 301
SegName:
Name: O3B
Index: 3842
Type: O3B
Value: 11.289

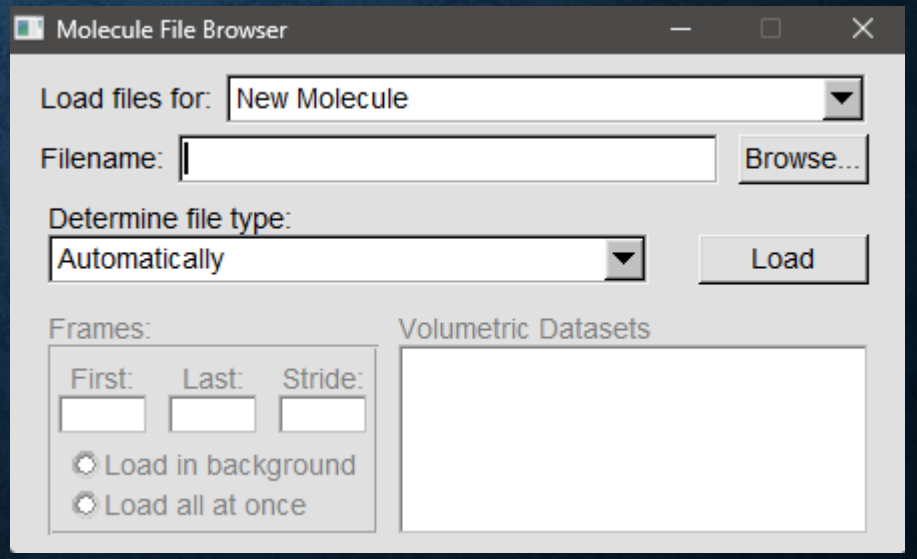
GETTING STARTED WITH VMD

- The Three Windows
  - **VMD Main:** The control hub. Use it to load files and manage molecules.
  - **VMD Display:** The 3D canvas. Where your protein comes to life.
  - **VMD Console:** For power users. Type commands directly here. [Note: Usually hidden by default]
- Mouse Controls
  - Left-click + Drag: 3D Rotation (Free rotation).
  - Right-click + Drag: Z-axis Rotation (Rotation in the projection plane).
  - Scroll Wheel: Zoom in / out.
- Essential Hotkeys
  - Keyboard "R/T/S"**:** Switch to **Rotate** / **Translate** / **Scale** mode.
  - Keyboard "=":


VMD 1.9.3 OpenGL Display
Display
VMD Main
File Molecule Graphics Display Mouse Extensions Help
ID T A D F Molecule Atoms Frames Vol
Main
zoom
Loop
step
1
speed
VMD TkConsole
File Console Edit Interp Prefs History Help
loading history file ... 48 events added
Main console display active (Tcl8.5.6 / Tk8.5.6)
(Desktop) 49 %
Console

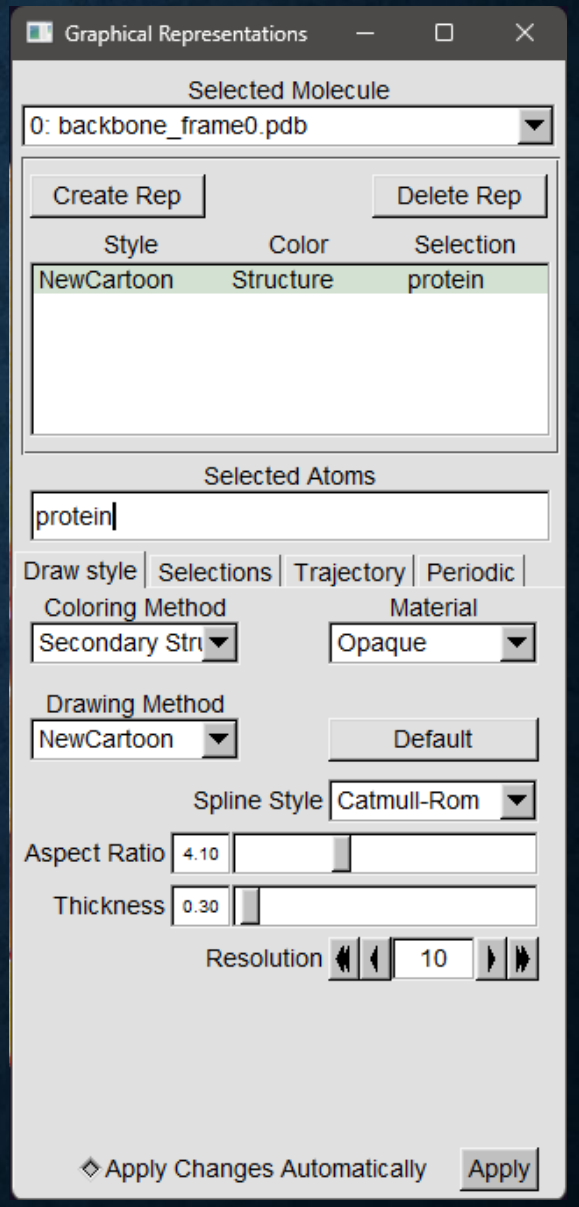
LOADING YOUR FIRST MOLECULE IN VMD

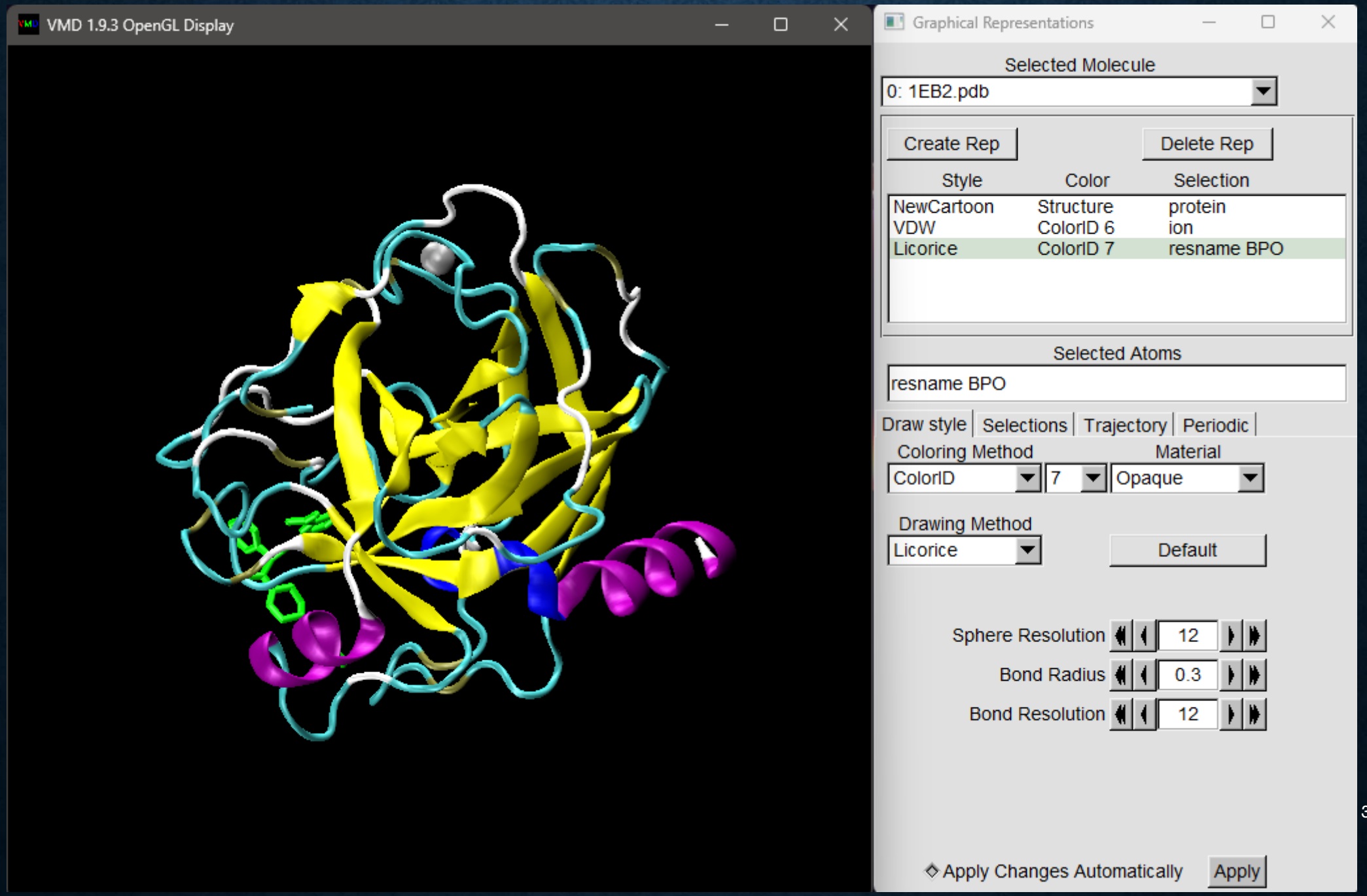
1. In the VMD Main window, go to "File" → "New Molecule".
2. Click "Browse" and select the 1EB2.pdb (or .cif) you just downloaded.
3. Ensure "Determine file type:" is set to "Automatic ".
4. Click "Load".



STRUCTURAL VISUALIZATION – CONFIGURING REPRESENTATIONS

1. Accessing Settings
   - Path: Navigate to "Graphics" → "Representations…" in the VMD Main window.
2. Defining Selection Range
   - Selected Atoms: Ensure the input box shows "protein" to apply changes to the protein.
3. Core Configuration Steps
   - Coloring Method: Select "Secondary Structure".
   - Drawing Method: Change from Lines to "NewCartoon".




VMD 1.9.3 OpenGL Display
Graphical Representations
Selected Molecule
0: 1EB2.pdb
Create Rep
Delete Rep
Style
Color
Selection
NewCartoon
Structure
protein
VDW
ColorID 6
ion
Licorice
ColorID 7
resname BPO
Selected Atoms
resname BPO
Draw style
Selections
Trajectory
Periodic
Coloring Method
Material
ColorID
7
Opaque
Drawing Method
Licorice
Default
Sphere Resolution
12
Bond Radius
0.3
Bond Resolution
12
Apply Changes Automatically
Apply

THE CHALLENGE: UNSOLVED STRUCTURES

- Homology Modeling
  - "Evolutionary Conservation." If two proteins share high sequence similarity, they likely share a similar 3D fold.
  - Workflow: Search PDB for a Template → Align sequences → Build 3D coordinates.
  - Common Portal: SWISS-MODEL (https://swissmodel.expasy.org/).
- Generative AI - AlphaFold 3
  - A generative AI model that uses Attention-based Pairformers to process molecular relationships and a Diffusion Module to directly predict all-atom 3D coordinates.
  - AlphaFold Protein Structure Database: Over 200 million predicted structures available for instant download.
  - AlphaFold Server: A Google-hosted environment to predict "unsolved" or custom sequences using high-performance GPUs.

AlphaFold Protein Structure Database (https://alphafold.ebi.ac.uk/)

AlphaFold Protein Structure Database

Home About FAQs Downloads API Collaborators

Search for protein, gene, UniProt accession or organism or sequence search

Search

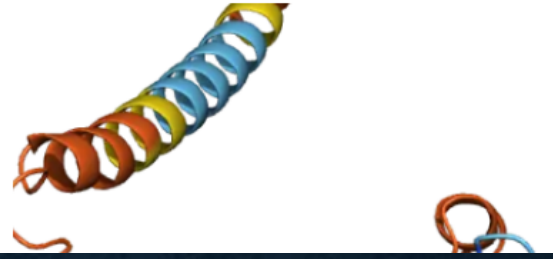
Examples: MENFQKVEKIGEGTYGV... Free fatty acid receptor 2 At1g58602 Q9I1F6 E. coli

See search help Go to online course

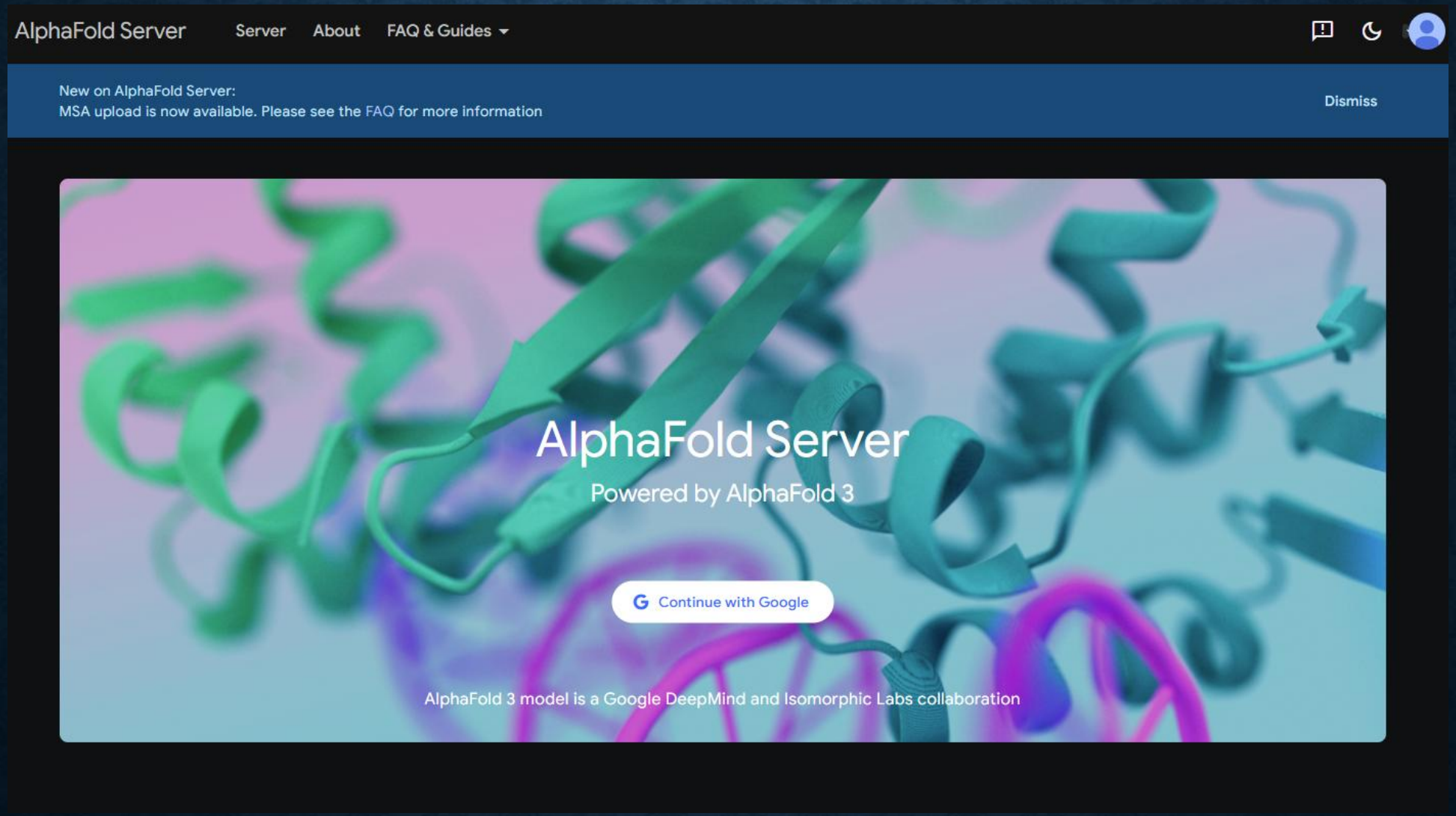
AlphaFold DB provides open access to over 200 million protein structure predictions to accelerate scientific research.

Background

AlphaFold is an AI system developed by Google DeepMind that predicts a protein's 3D structure from its amino acid sequence. It regularly achieves accuracy competitive with



AlphaFold Server (https://alphafoldserver.com/welcome)



PREDICTING PROTEIN STRUCTURES

- Try the Classical Approach (SWISS-MODEL)
  - Go to SWISS-MODEL (https://swissmodel.expasy.org/).
  - Click "Start Modelling" and paste your sequence.
  - Mechanism: It will search for a Template (a similar protein) to "copy" the fold.
  - Observe: Check the Global Model Quality Estimate (GMQE) score.
- Try the Next-Gen AI (AlphaFold 3)
  - Go to AlphaFold Server (https://alphafoldserver.com/welcome).
  - Sign in (Google account required) and paste your sequence.
  - Mechanism: It uses Generative Diffusion to predict the structure from scratch, even without a clear template.
  - Observe: Check the pLDDT confidence scores (displayed in colors).

Serine protease 1 (.fasta format)

>sp|P00760|TRY1_BOVIN Serine protease 1 OS=Bos taurus OX=9913 GN=PRSS1 PE=1 SV=3

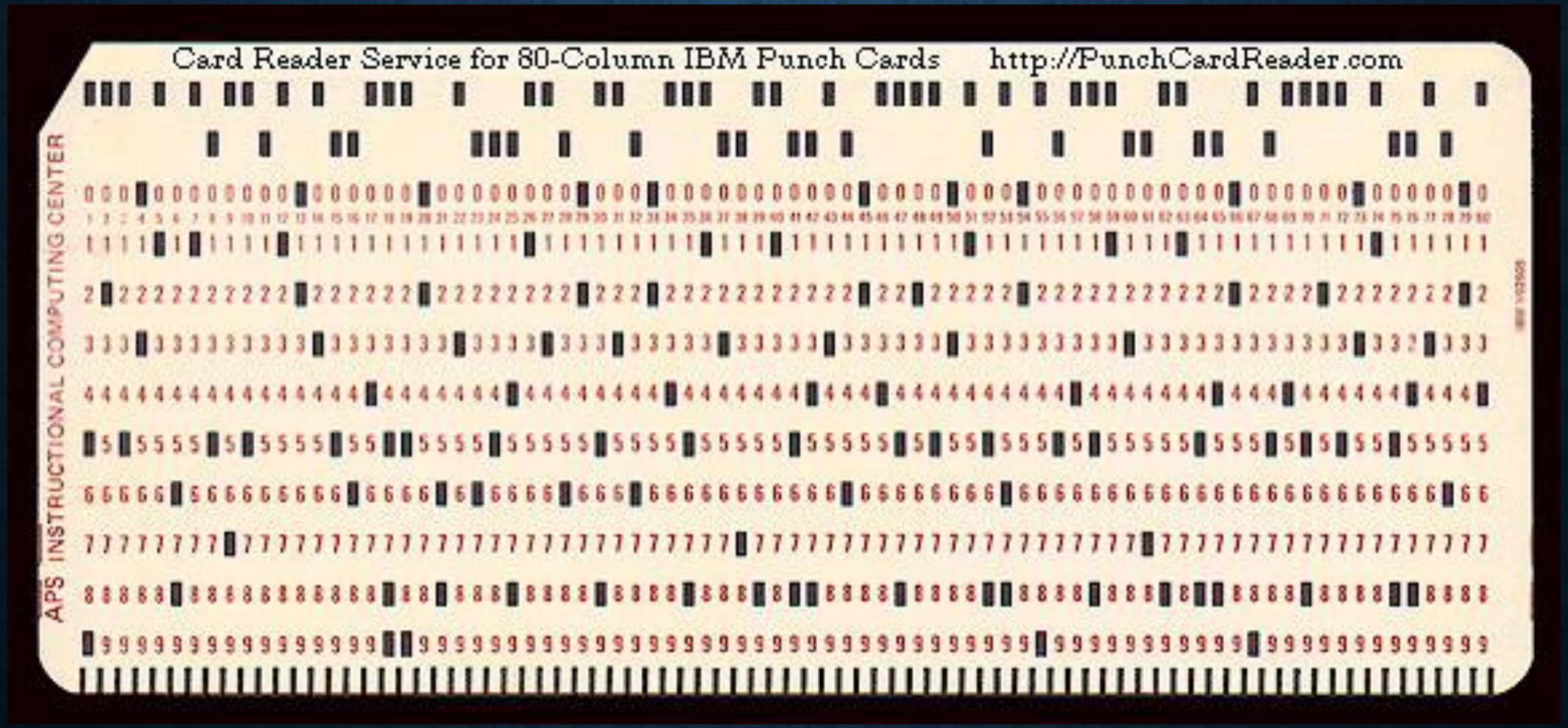
MKTFIFLALLGAAVAFPVDDDDKIVGGYTCGANTVPYQVSLNSGYHFCGG
SLINSQWVVSAAHCYKSGIQVRLGEDNINVVEGNEQFISASKSIVHPSYN
SNTLNNDIMLIKLKSAASLNSRVASISLPTSCASAGTQCLISGWGNTKSS
GTSYPDVLKCLKAPILSDSSCKSAYPGQITSNMFCAGYLEGGKDSCQGDS
GGPVVCSGKLQGIVSWGSGCAQKNKPGVYTKVCNYVSWIKQTIASN

- Red – Signal
- Yellow – Propeptide
- White – Main Chain

SUPPLEMENTARY INFORMATION

EVOLUTION OF DATA FORMATS: PDB VS. PDBx/mmCIF

- The Classic PDB Format (.pdb)
  - Heritage: A legacy format since the 1970s, based on 80-column punched cards.
  - Limitations:
    - **Capacity:** Can only hold up to **99,999 atoms** and **62 chains**.
      (Certain software use Hexadecimal to extend this, but it often causes compatibility errors across different platforms.)
    - **Ambiguity:** Limited column width leads to loss of precision for very large coordinates.
- The New Standard: PDBx/mmCIF (.cif)
  - Future-Proof: The official format of wwPDB since 2014; mandatory for all new depositions.
  - Advantages:
    - Unlimited Scale: No limit on the number of atoms or chains (Essential for Ribosomes and Viruses).
    - Rich Metadata: Contains much more detailed experimental information and chemical descriptions.


Card Reader Service for 80-Column IBM Punch Cards
http://PunchCardReader.com
APS INSTRUCTIONAL COMPUTING CENTER

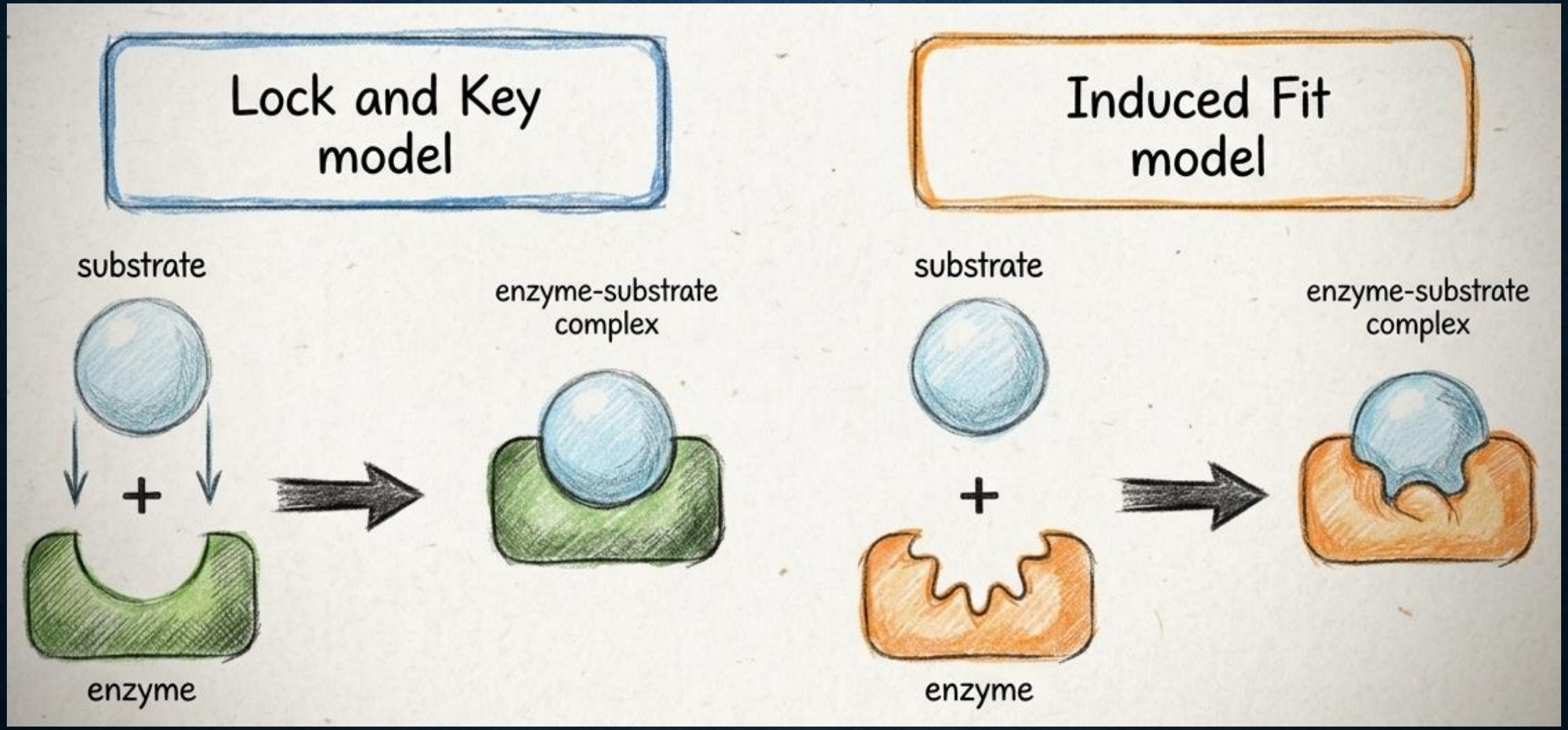
IS NEWER ALWAYS BETTER? (THE mmCIF VS. PDB DILEMMA)

- The Standard vs. The Toolchain
  - The "New" (mmCIF): Technically superior, unlimited capacity, the official gold standard for archiving.
  - The "Old" (PDB): Riddled with limitations (99k atoms), but deeply embedded in 40 years of scientific code.
- Why the Industry Stays "Old“
  - Legacy Code Burden: Many core MD engines (NAMD, GROMACS) use 5-digit integer indexing. Changing this requires a fundamental rewrite of the low-level source code.
  - The "Broken" Dependency Chain: MD workflows involve many interconnected tools (Topology builders, solvation scripts). If one tool switches to mmCIF but others don't, the entire research pipeline fails.
- The Practical Compromise
  - Workaround: Instead of moving to mmCIF, developers use Hexadecimal or Hybrid-36 to "stretch" the old PDB format.
  - **The Risk:** This creates "non-standard" files that may work in VMD but cause errors when uploaded to web servers or read by other software.

FROM STRUCTURE TO INTERACTION MOLECULAR DOCKING PREDICTION

WHAT IS MOLECULAR DOCKING

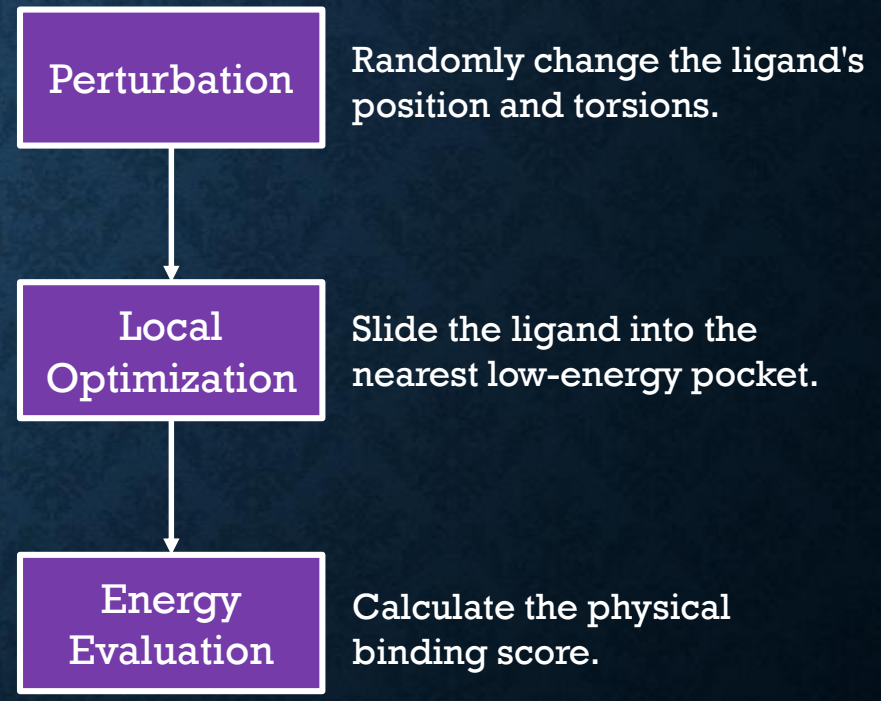
- Core Concepts
  - Computational simulation of a small molecule (ligand) binding to a protein (receptor).
  - Two Goals:
    1. Predict Binding Pose: Where does it dock? How does it fit?
    2. Predict Binding Affinity: How tightly do they bind?
- Computational Logic
  - Step 1 – Sampling: Translating, rotating, and folding the ligand to explore 3D conformations.
  - Step 2 – Scoring: Calculating physical energies (H-bonds, Van der Waals, Electrostatics).
- Theoretical Models
  - Lock & Key Model: Assumes the protein is a Rigid Body. Fast calculation, common in basic docking.
  - Induced Fit Model: Protein acts like a Glove, changing its conformation slightly to adapt to the drug.


Lock and Key model
substrate
enzyme-substrate complex
enzyme
Induced Fit model
substrate
enzyme-substrate complex
enzyme

AUTODOCK VINA
THE STANDARD ENGINE

The Benchmark of Molecular Docking

- Global Standard: One of the most widely cited and utilized open-source docking engines globally.
- Why Vina? Offers a massive leap in speed and accuracy over legacy programs by treating docking as a global optimization problem.



PREPARATIONS (EXAMPLE: LOCK & KEY MODEL)

- Ligand
  1. Identify the rigid core of the molecule.
  2. Define which chemical bonds can freely rotate.
  3. Remove apolar hydrogens.
  4. Save as "ligand.pdbqt".
- Receptor
  1. Remove crystallized water molecules.
  2. Remove unnecessary components (ions or other small molecules co-crystallized).
  3. Add polar hydrogens (for hydrogen bond calculations).
  4. Save as "receptor.pdbqt".
- Defining the Grid box
  1. Define box center: Where is the bonding pocket?
  2. Define box size: How large is search area?

  Note: A larger box takes exponentially more time to calculate.

EVOLUTION OF DOCKING TOOLS

Classic Era (2010-2020)

- MGLTools + AutoDock Vina 1.1.x

[+] Intuitive GUI: Excellent for visual learning and manual Grid Box setup.

[–] Legacy Architecture: Obsolete Python 2. No native Apple Silicon support.

[–] UI Instability: Prone to graphical glitches and crashes on modern OS.

[–] Bottleneck: Manual clicking is unscalable for large datasets.

Modern Era (2021-Present)

- Meeko + AutoDock Vina 1.2.x

[+] Official Standard: The currently recommended engine by AutoDock.

[+] Modern Tech Stack: Python 3 & CLI-based. Importable into custom scripts.

[+] Built for HTVS: Integrates with RDKit for High-Throughput Virtual Screening.

[–] Steep Learning Curve: No GUI ("blind prep"). Relies on tools like OpenBabel for format conversion.

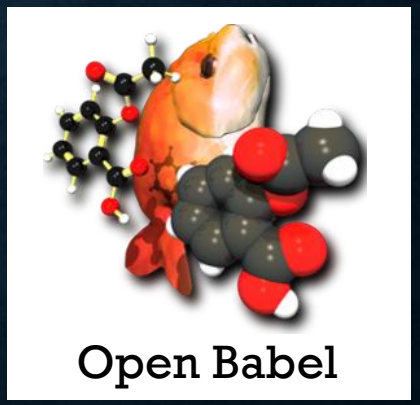
SOFTWARE DOWNLOADS

- AutoDock Vina
  - https://vina.scripps.edu/
- MGLTools
  - https://ccsb.scripps.edu/mgltools/
- Meeko
  - https://meeko.readthedocs.io/en/develop/
- RDKit
  - https://www.rdkit.org/
- Open Babel
  - https://openbabel.org/index.html



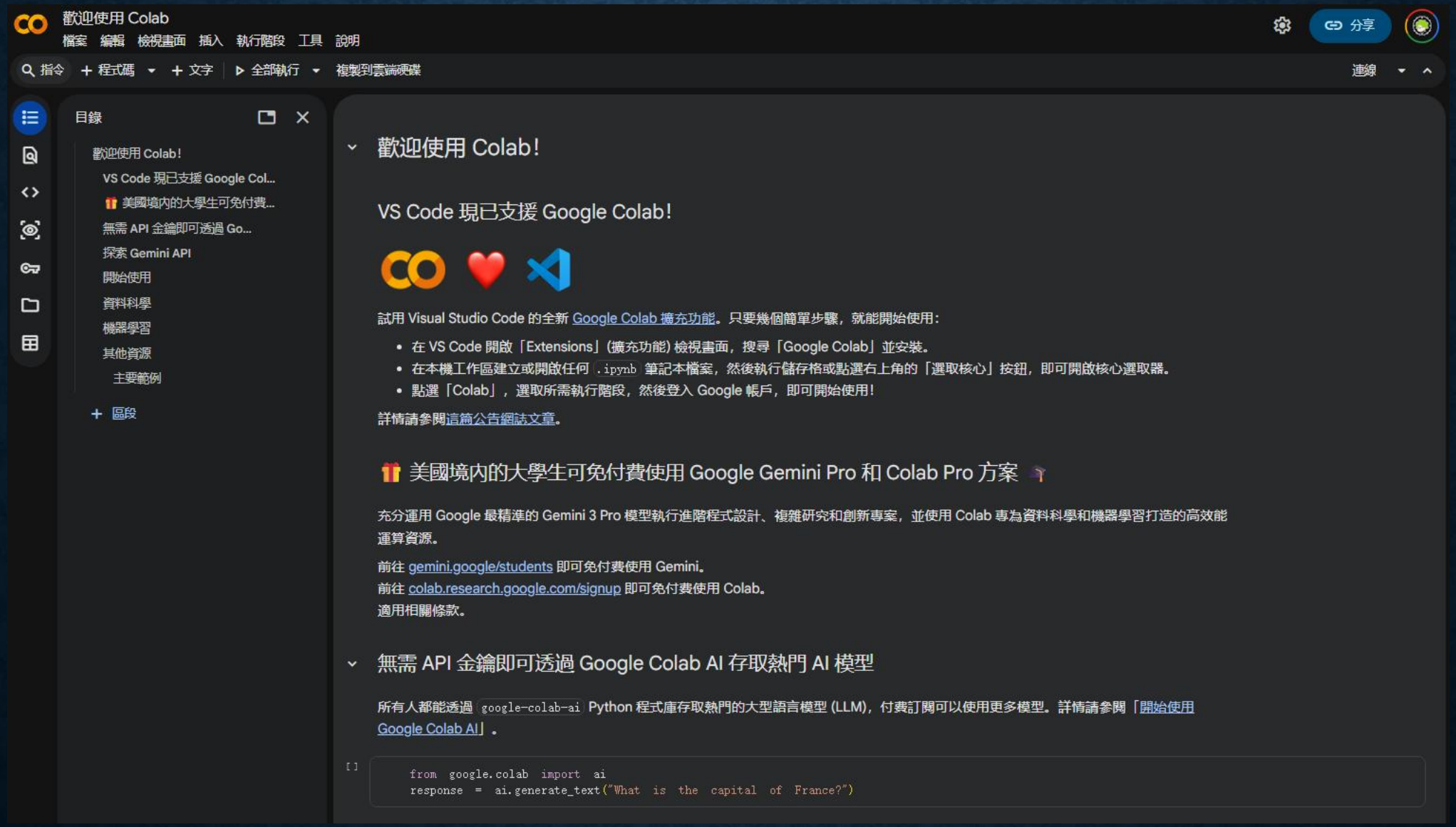






CLOUD-BASED TEACHING FRAMEWORK

- Overcoming Hardware Barriers
  - Bypasses local installation issues and OS incompatibilities (such as Apple Silicon limits) by moving the entire workflow to the cloud.
- Google Colab (Powered by Jupyter)
  - Provides free, ready-to-use cloud servers.
  - Users only need a web browser to access powerful Python computing resources with zero setup.
- Interactive & Transparent Learning
  - Integrates Meeko and Vina into a seamless GUI.
  - Users can run docking simulations instantly, while still having the freedom to inspect the underlying Python code.


歡迎使用 Colab
檔案 編輯 檢視畫面 插入 執行階段 工具 說明
分享
指令 + 程式碼 + 文字 全部執行 複製到雲端硬碟 連線
目錄
歡迎使用 Colab!
VS Code 現已支援 Google Col...
美國境內的大學生可免付費...
無需 API 金鑰即可透過 Go...
探索 Gemini API
開始使用
資料科學
機器學習
其他資源
主要範例
+ 區段
歡迎使用 Colab!
VS Code 現已支援 Google Colab!
試用 Visual Studio Code 的全新 Google Colab 擴充功能。只要幾個簡單步驟，就能開始使用：
在 VS Code 開啟「Extensions」(擴充功能) 檢視畫面，搜尋「Google Colab」並安裝。
在本機工作區建立或開啟任何 .ipynb 筆記本檔案，然後執行儲存格或點選右上角的「選取核心」按鈕，即可開啟核心選取器。
點選「Colab」，選取所需執行階段，然後登入 Google 帳戶，即可開始使用！
詳情請參閱這篇公告網誌文章。
美國境內的大學生可免付費使用 Google Gemini Pro 和 Colab Pro 方案
充分運用 Google 最精準的 Gemini 3 Pro 模型執行進階程式設計、複雜研究和創新專案，並使用 Colab 專為資料科學和機器學習打造的高效能運算資源。
前往 gemini.google/students 即可免付費使用 Gemini。
前往 colab.research.google.com/signup 即可免付費使用 Colab。
適用相關條款。
無需 API 金鑰即可透過 Google Colab AI 存取熱門 AI 模型
所有人都能透過 google-colab-ai Python 程式庫存取熱門的大型語言模型 (LLM)，付費訂閱可以使用更多模型。詳情請參閱「開始使用 Google Colab AI」。
[ ]
from google.colab import ai
response = ai.generate_text("What is the capital of France?")

COLAB ENVIRONMENT & SYNTAX GUIDE

1. The Cloud Environment
   - What is it? You are borrowing a Google Linux server.
   - Benefit: Zero local installation. Runs entirely in your browser.
2. Syntax Basics
   - Regular Text: Standard Python 3 code.
   - # (Comment): Notes for humans. Ignored by the system.
   - ! (Shell Command): Runs a command on the underlying Linux OS.
   - % (Magic Command): Special shortcuts for the Jupyter environment.
3. Software Installation
   - !pip install (Python): Installs Python modules (e.g., vina, rdkit).
   - !apt-get install (Linux OS): Installs system-level software (e.g., openbabel).

COMPUTATIONAL TOOLKIT

- Docking Engine
  - vina: The native Python API for executing AutoDock Vina directly.
- Structure Preparation
  - meeko: Automated receptor and ligand preparations.
  - `rdkit` & openbabel: Chemoinformatics engines for 3D conformation and format conversion.
  - Bio.PDB/gemmi/prody: Parsing, filtering, and cleaning complex protein structures.
- Visualization & UI
  - py3Dmol: In-browser 3D rendering for structural inspection.
  - ipywidgets: Interactive GUI elements (buttons, sliders) for a user-friendly experience.
- Data Pipeline
  - numpy: Spatial matrix calculations (e.g., Grid Box auto-centering).
  - colab.files & zipfile: Managing file uploads, downloads, and workspace cleanup.

MOLECULAR DOCKING TOOLBOX

A Comprehensive Framework for Lead Optimization and High-Throughput Screening

- Vina-Vision

1. Interactive Pose Refinement
2. Atom-Name Metadata Retention
3. 3D Visual Box Positioning

https://github.com/ycliu0214/MD-CADD-Courseware/blob/main/Jupyter-Notebook/AutoDock_Vina_Workflow.ipynb

- Vina-HTVS

1. GPU-Accelerated High-Throughput Virtual Screening (HTVS)
2. Similarity Expansion Search
3. Dynamic Centroid Calibration

https://github.com/ycliu0214/MD-CADD-Courseware/blob/main/Jupyter-Notebook/AutoDock_Vina_GPU_Workflow.ipynb

AutoDock_Vina_Workflow.ipynb

檔案 編輯 檢視畫面 插入 執行階段 工具 說明

指令 + 程式碼 + 文字 全部執行 連線

CPU 精準視覺化對接 (Interactive Precision Docking)

專為「先導藥物優化 (Lead Optimization)」與「結合機制研究」設計。 強調100%的使用者掌控度、即時的3D視覺互動，以及最高精度的單一分子對接演算。

環境建置與核心套件載入

功能說明：一鍵安裝對接與視覺化所需的所有底層套件（包含 AutoDock Vina CPU 引擎、py3Dmol 3D 渲染引擎、RDKit 與 OpenBabel）。

核心設計：採用靜默安裝 (-q) 保持版面整潔，並確保所有化學資訊學套件的版本相容性，為後續的 3D 運算打下基礎。

```
# 隱藏安裝輸出來保持版面整潔
!pip install -q meeko vina py3Dmol ipywidgets biopython rdkit gemmi prody
!apt-get install -q -y openbabel
```

顯示隱藏的輸出內容

```
# 系統環境與檔案讀寫 (System & I/O)
import os
import sys
import time
import zipfile
import glob
import subprocess
from google.colab import files   # Colab 專屬檔案上傳/下載

# 數值運算與資料解析 (Math & Data Parsing)
import math                      # 用於處理比對問題
import re                        # 用於解析 Vina 的文字報告
import numpy as np               # 用於計算蛋白質質心與網格邊界
from datetime import datetime    # 用於產生打包檔案的時間標記

# 互動介面與視覺化 (UI & Visualization)
import ipywidgets as widgets                    # UI 滑桿、按鈕與下拉選單
```

GPU 高通量虛擬篩選 (High-Throughput Virtual Screening)

專為「藥物庫擴展」與「新穎骨架發現」設計。利用 PubChem 巨量資料庫與 GPU 叢集的強大平行運算能力，將常規的篩選時程由數天大幅壓縮至數小時內。

Vina-GPU 的突破性改良

Vina-GPU並非由美國斯克里普斯研究所 (The Scripps Research Institute, TSRI) 的分子圖學實驗室 (Molecular Graphics Laboratory)開發，而是由南京郵電大學 (NJUPT) 的 Delta Group 聯合其他運算實驗室進行拓展。

它將最耗時的「評分函數計算」與「梯度幾何優化」演算法進行了底層重寫，封裝成 OpenCL 與 CUDA 的核心指令 (Kernels)。這項架構替換使得：

1. 千倍並行計算：原本由 CPU 依序處理的幾千個初始隨機配體姿勢 (Random Poses)，現在可以直接分配給 GPU 上的數千個流處理器 (CUDA Cores) 「同時」 進行局部最佳化與能量算分。
2. 網格數據常駐：將受體的 Grid Map 數據快取於 GPU 的高頻寬記憶體 (VRAM) 中，徹底打破了傳統 CPU 架構下的記憶體存取瓶頸 (Memory Bandwidth Bottleneck)，實現真正的高通量。

由於部分演算法的替換，雖然其能量估算仍採用Vina的物理函式，但其結果仍與CPU運算的Vina具有差異。

Vina-GPU 引擎編譯與環境自適應

功能說明：偵測雲端 GPU 硬體拓樸（如 T4, V100, A100），動態修改 Makefile 進行 Vina-GPU 2.1 引擎的 C++ 與 OpenCL 底層編譯。

核心設計：

1. 雲端熱備份機制 (Hot-Backup)：首次成功編譯後自動鏡像至 Google Drive。後續執行若偵測到同構型硬體，將直接觸發「秒速喚醒」，免除重複編譯的效能損耗。
2. 硬體架構防呆鎖 (Architecture Interlock)：若系統偵測為純 CPU 執行階段，將觸發 RuntimeError 強制攔截，並提供標準的環境切換指引，避免高通量管線崩潰。

不過由於一些相依的函式庫暫時無法靜態編譯與備份，每次開啟新的工作階段必須要重新進行下載安裝與編譯。上述步驟都已自動化降低使用

DEMO TARGETS

- Receptor
  - Target: 1EB2 (RCSB PDB)
  - Action: Download "Legacy PDB Format“
  - Link: https://www.rcsb.org/structure/1EB2
- Ligand
  - Target: BPO
  - Action: Download "3D Conformer" (SDF format)
  - Link: https://pubchem.ncbi.nlm.nih.gov/compound/5287836